Large Language Model based Multi-Agent Systems: Architectures, Collective Intelligence, and Emerging Frontiers

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Contents

1 Introduction

The evolution of intelligent systems has traversed a remarkable trajectory, from classical multi-agent frameworks governed by predefined state-action spaces and explicit coordination protocols [1, 2, 3] to the contemporary paradigm of large language model (LLM) based multi-agent systems. This transition represents not merely a technological upgrade but a fundamental shift in how we conceptualize agency, coordination, and collective intelligence. Classical multi-agent systems, despite their mathematical elegance and formal guarantees [4, 5], were constrained by rigid pre-specification and the necessity of designing interaction protocols from first principles [6]. The advent of LLMs has dismantled these barriers, offering a new substrate where agents are no longer bound to finite action repertoires but operate within a rich, open-ended space of language-mediated reasoning, planning, and social interaction [7, 8, 9].

The motivation for orchestrating multiple LLM-based agents, rather than relying on a monolithic model, arises from fundamental limitations of single LLMs. A single model, despite its power, operates within a finite context window, is susceptible to individual hallucinations, and lacks the benefits of collaborative knowledge and specialized reasoning [10, 11]. By decomposing complex tasks and allocating subtasks among specialized agents with distinct roles and perspectives [12, 13], an LLM-based multi-agent system (LLM-MAS) can overcome these singular limitations. This collaborative division yields a collective that is far greater than the sum of its parts, enabling workflows that tackle problems involving layered reasoning, cross-disciplinary synthesis, and long-horizon planning [14, 15]. It is this synergistic utilization of their inherent reasoning, planning, in-context learning, and tool-use capabilities that defines the unique capability of such systems [16, 17].

What fundamentally distinguishes an LLM-MAS from either a single agent or a simple ensemble of models is a suite of core system-level properties. These include the presence of distinct, specified roles for agents, the use of natural language as the primary mode of flexible interaction, and the capacity for emergent collective behavior that superlinearizes individual capabilities [18, 19]. These systems are not just about combining outputs; they are about creating a dynamic social architecture where information flows, continuous and conflicting viewpoints are negotiated [10], and a "society of minds" is implemented [20]. This interaction yields identifiable social and behavioral patterns, from conformity to the emergence of trust, reflecting the social psychology of the agents, albeit grounded on artificial substrates [21]. Furthermore these architectures facilitate dynamic auto-organization—without a single orchestrator guiding the processes—which is the sine qua non for achieving scalable, generalist intelligence [22, 23].

The urgency and importance of this survey stem from the fragmented narrative that currently defines the field. Existing examinations often articulate only a narrow component of the multi-agent experience, such as the nature of agent personas, the mechanics of the memory module [24], or the specifics of planning schemes [25]. Others focus on benchmark-specific advances, like those in a particular professional domain such as software engineering [26], often missing the underlying system-level principles that determine success. There is a definitive gap in synthesizing its architecture, the subtle communication and coordination mechanisms, and the emergent group dynamics into a single coherent science [27]. Our work aims to fill this void by introducing a comprehensive and unified framework that links system architecture, group dynamics, applications, and the profound societal implications of deploying these cognitive artifacts [28]. We address the central engineering challenges of orchestration and adaptive dynamics, while simultaneously providing a critical perspective on the fundamental boundaries of structural capabilities and validation. The promising—and often destabilizing—potential of the society of LLMs is analyzed through the lens of systematic evaluation of emergent properties, and an open security paradigm. This positions the survey as a foundation for subsequent research and a practical guide for navigating this complex terrain, from the nature of human-machine collective intelligence to the pragmatic buildout of large-scale, ever-evolving agent ecosystems.

2 Architectural Design of Agents and Multi-Agent Systems

2.1 Agent Profiling and Persona Engineering

Agent profiling constitutes the foundational design layer in LLM-based multi-agent systems, determining how individual agents perceive their identity, interpret their responsibilities, and interact with both peers and tasks. The profiling process has evolved from simple role designation to a sophisticated engineering discipline that draws upon social psychology, organizational theory, and cognitive science, directly shaping the division of cognitive labor that distinguishes a true multi-agent system from a monolithic model [27].

The most prevalent approach to role specification leverages natural language prompting, where detailed system messages define professional personas such as software engineers, financial analysts, or legal advisors, encoding both domain expertise and procedural knowledge. A notable case involves courtroom simulation frameworks, which assign distinct professional identities that carry implicit behavioral protocols and communication standards [12]. Empirical assessments of such role assignments demonstrate that persona-conditioned agents display higher reasoning capabilities on capability-appropriate tasks. This specialization arises because natural-language role definitions shape output distributions toward expected expertise patterns and help agents select appropriate reasoning strategies. A key design decision in the role specification is the granularity of role definitions: broad characterizations may lead to inconsistent escalating population behavior, while behavioral protocols and signaling lower possible durations create oppression and inflexible liability. Advanced frameworks address this through checklists, specialized and planning-based interaction strategies, and structured communication protocols that mitigate cascading failures due to role conflicts [21] [22]. Sensible forecasting tools grounded in role-specific (procedural, declarative, episodic) knowledge loops make memory multiple anchors for behavior and identity [29, 24].

For role definition to anchor and stabilize behaviors, agents may be enhanced with dynamically defined roles, automatically constructed or strategically selected based on analysis of past experience and current context [22]. Studies show that role-based hierarchies—in particular, involving designated leadership and novel organizations—can reduce communication costs and improve team efficiency [30]. Some frameworks employ extensive behavioral variability and function-grained instructions to ensure the domain expert roles remain consistent with the in-task demands [31].

Additionally, rich cognitive architectures rely on perceived bodies or emotion models to reveal human traits, such as risk attitude, cooperativeness, or conformity. Such individualizations introduce heterogeneity into team reasoning, enabling emergent coordination and division of labor [21]. Social simulation with role-specific personas (e.g., different value stances, information-processing styles) may inadvertently stabilize individual or collective behavior, making cooperation or debate collapse into consensus [31]. In multiple agents end-to-end, over time the use of profiles, or role drift or conformity, can strip the initial diversity. For this reason, personas can be dynamically adjusted based on behavior evolution and responsibility change [22]. Thus agent profile engineering is not simply setting roles in a prompt, but as integration with underlying system architectures such as memory and feedback that allow a robust multi-agent composition [27]. This trend points toward autonomously generating and advancing agents for performing increasingly complex collective tasks.

2.2 Agent Memory Architectures and Knowledge Management

Memory architectures constitute the substrate upon which both individual agent cognition and collective intelligence are built, transforming the stateless role-playing defined by profiling into temporally-aware systems capable of learning from experience and coordinating over extended horizons. If profiling determines *who* agents are, memory determines *how they persist*—how they remember, learn, and adapt the persona assigned to them. Drawing upon cognitive science, contemporary designs partition memory into functionally distinct components that operate across different timescales, with the choice of data structures, retrieval mechanisms, and integration strategies fundamentally shaping the capability ceilings of the overall system. At the individual level, the predominant framework distinguishes between short-term working memory, which holds immediate context and intermediate reasoning states during a single episode, and long-term memory, which is in turn subdivided into episodic memory (a record of past experiences) and semantic memory (structured world knowledge and learned abstractions). This cognitive scaffolding can be traced directly to classical agent architectures that first formalized the relationship between belief, desire, and intention, but LLM-based systems increasingly depart from these rigid formalisms by implementing these components as dynamic, natural language-mediated stores that interleave with the agent's planning and decision-making loop [24]. Crucially, the profiles of agents—including their roles, personas, and behavioral protocols—are not static declarations but are stored, retrieved, and constantly reshaped through the memory system. A role definition is an episodic record periodically activated, refined, or discarded, and the identity an agent projects is a combination of its current role prompt and the cumulative traces of past interactions, preserved and distilled into abiding behavioral dispositions.

A fundamental tension in memory design concerns retrieval. The capacity limitations of finite context windows make the naive strategy of concatenating all historical interactions computationally expensive and detrimental to performance, as irrelevant details dilute attention and degrade the coherence of the autoregressive generation process. Consequently, targeted retrieval has become the dominant approach, wherein agents choose relevant embeddings of current states and their semantic similarity to stored memories, combining dense vector retrieval with structured graph traversals to balance precision and recall and facilitate associative, non-linear recall across distinct but related experiences. Reflection, which reinstates higher-order inferences, self-critiques, and behavioral policy adjustments from first-order observations, provides a mechanism for consolidating episodic records into structured semantic knowledge. This iterative deepening from raw events into abstract, verifiable representations mitigates the risk of memory decay and ensures that the accumulated wisdom of past interactions informs future action, allowing each agent-character to refine its own behavioral policy without catastrophic forgetting. The recall mechanism determines not only which memories are accessed but also *which aspects of an agent's acquired role and identity are reactivated*, so that the same objective event may be filtered and interpreted differently depending on the stored, historically-conditioned persona profiles. Recent advances have further endowed agents with self-adaptive capabilities, where the system monitors its own performance to trigger structural and behavioral changes in response to dynamic environments, echoing the adaptive profiling discussed earlier.

In multi-agent systems, however, the most defining shift in memory design is its expansion from a private, internal module to a shared, relational repository that coordinates childbirth—the group-level cognition. In these collective constructions, the memory space transcends the boundaries of individual agents, evolving into an externalized semantic memory substrate that all participating agents can read from and write to. This shared repository occupies a central role in governing inter-agent dynamics: it tracks the group's overall progress by continuously updating states of task decomposition, flagging completed subtasks, and identifying redundancies within agent workflows, thereby avoiding duplicated effort and ensuring a coherent autonomy between agents with functional roles. Whereas role profiles define *who* each agent is, and architecture defines the procedures *by which* they connect, this shared memory defines *what they collectively know and remember*—the group-level repository that preserves not only the tasks, but also the negotiated norms, by observing the behaviors of other agents, and the emergent social structures that arise as coordination unfolds. Because multi-agent LLM systems are by necessity societies of mind, where communication relies on language exchanges, the shared memory architecture functions as a group-level relational database that extends beyond the finite context constraints of any single actor. This framework enables asynchronous collaboration and supports a persistent, evolving digital institutional space, where peer agents provide feedback and substantive critique, prompting reflection and realistic narrative construction rather than accumulating disjointed facts.

Nevertheless, aligning individual privacy with group-level collective insights remains an open and crucial challenge. This unresolved tension is compounded by the fact that role specialization in agent profiling often conflicts with the sharing primitive required for effective group memory: agents must strategically share knowledge to remain coherent, yet they must also maintain the value and purpose of their differentiation. It requires them to coordinate selective information deployment, to analyze the dynamic flow of the evolving communication graph, and to ensure a governance structure that expresses the effective integration of high-integrity temporal knowledge despite the continuous emergence and obsolescence of new facts. The field is thus moving away from static memory modules as passive stores, toward memory systems as dynamic, learnable components of the agent network. Successful architectures must integrate encoding, consolidation, and retrieval stages as a unifying end-to-end system capable of pruning and prioritizing iterative information, and semantically differentiating between private vs. public memories based on role relevance and group needs. This dynamic gating enables the profiling layer—discussed earlier—to remain responsive to experience, and also paves the way for the architectural layer—discussed next—that will determine how information flows through the optimized group structure. The future direction of this subsystem lies in the joint optimization of individual and group memory pathways, where quantifiable agent importance metrics might guide the self-refinement of team composition, and where "knowledge management" evolves from a fixed resource into a living, collective conscience of the shared identity and evolution of the system.

2.3 System Architectures and Organizational Topologies

The architecture of a multi-agent system—the topological arrangement of its constituent agents and the resulting pathways for control, data, and influence—is a primary determinant of its collective capabilities, operational efficiency, and robustness. These structural configurations define the fundamental character of the system, shaping how tasks are decomposed, how information propagates, and how conflicts are resolved. A systematic understanding of these canonical organizational topologies, and their associated trade-offs, is therefore essential for the principled design of effective and resilient systems.

The most prevalent architecture for orchestrating LLM-based agents is **centralized or hierarchical coordination**. In this hub-and-spoke model, a single orchestrator or lead agent acts as the central planner and task allocator, decomposing high-level objectives into sub-tasks that are distributed to specialized agent workers. This design, exemplified by the "Hierarchical Auto-Organizing System" [15] and frameworks like CoELA [13], employs a dual-pronged structure ensuring centralized planning and decentralized execution. The primary advantage is the close alignment of the system to the task's hierarchical nature and the ability for the central planner to maintain a clear global view, enabling efficient task management. However, this centralization introduces a **single point of failure** and a critical bottleneck: the orchestrator's performance is constrained by its finite context window and reasoning capacity. As the number of agents or the complexity of tasks scale beyond its capacity, the coordinator's ability to track status and make decisions degrades, fundamentally limiting the system's scalability [18]. This architectural pattern is archetypal of hierarchical leadership in teams, an inspiration for new organizational prompt designs in LLM agents [30], and is used in event-driven manager-agents in driving scenarios [32].

In contrast to the central control of hierarchical models, **decentralized and flat organizational models** distribute authority across peer agents that interact to achieve a shared goal without a central authority. These configurations exhibit high resilience and fault tolerance, as the failure of one node does not cripple the system. In these peer-to-peer arrangements, agents coordinate via direct communication and self-organize, adapting their interaction topology to suit the task needs. For instance, in long-horizon cooperation, agents can dynamically form teams based on task requirements to navigate the environment or generate large-scale codebases [18]. The complexity in such systems quickly arises from the need for consensus; without centralized coordination, decision-making requires robust negotiation and voting mechanisms that arise from specific organizational structures [33]. This often leads to significant **communication overhead**, where the cost of negotiation and information exchange scales quadratically with the number of agents, a critical limitation for very large populations [34]. The choice between a centralized planner and a decentralized swarm is fundamentally a trade-off between control and per-agent autonomy.

Building upon these static configurations, the field is increasingly moving toward **dynamic reconfiguration and self-organization**, where the network topology is not predefined but adapts during task execution. This allows systems to maintain both the coherence of hierarchical control and the flexibility of decentralized structures. For instance, auto-organizing intra-communication mechanisms demonstrate dynamic group adjustments, where coordination hierarchy emerges from and changes based on the subtasks [15]. Such self-organizing systems, which can dynamically add or remove agents or shift leadership, are far more robust to environmental changes and context. This does, however, introduce a new layer of complexity, as the system must now handle the meta-coordination required to define its own dynamic structure, often without a predefined set of Standard Operating Procedures (SOPs) [18].

The choice of topology is mediated by an encoded set of trade-offs that span multiple dimensions, and formalizing this process is challenging. However, recent works have begun to explore these dynamics in a formal way. The relationship between these dimensions can be viewed as a multi-objective optimization problem, where the goal is to maximize the utility of the system measured by factors like task completion while minimizing coordination cost and latency. In practice, a designer must select a topology that balances these often-opposing forces. Studies have shown that while deep hierarchical collaboration provides the most robust reasoning , it is not always the most optimal format for complex and loosely coupled tasks, suggesting that the choice between a hierarchical reply, a flat debate, or a single agent is task-dependent [35]. The design of these architectures is further complicated by the operational context, which includes potential adversarial interactions and the requirement for interpretability. The challenge of orchestrating these agents becomes even more complex when we consider the diverse modalities required for real-world tasks and the inherent limitations of LLMs, requiring formal frameworks to map the design space and identify Pareto-optimal architectures. This nuanced understanding is a crucial prerequisite for developing robust systems that can leverage the capabilities of LLMs without succumbing to their architectural constraints [36].

2.4 Integration with Tools, Modular Frameworks, and External Environments

The operational efficacy of LLM-based multi-agent systems is fundamentally contingent upon their integration with external tools, modular frameworks, and heterogeneous computational environments. This integration layer determines whether an agent collective remains a closed text-generation artifact or evolves into a functional, deployable system capable of interacting with the physical and digital world. Conceptually, this integration can be situated within the broader abstraction of an "operating system" for agents, analogous to the role of a traditional OS in managing hardware and software resources [37]. This analogy is particularly salient for multi-agent systems, where resource contention, scheduling, and inter-process communication are not merely optimization problems, but fundamental requirements for coordinated behavior. Just as the architectural choices of topology shape the internal flow of control and information among agents, the integration layer defines the boundaries of that system—the interfaces through which the collective perceives, acts upon, and receives feedback from the world. The interrogative structure of the previous discussion does not serve as a standalone configuration but are realized semantically through this external integration environment.

A primary architectural concern is tool integration and the design of Application Programming Interfaces (APIs). Modern agent frameworks adopt an API-first design principle, prioritizing programmatic interfaces over user-centric UI interactions to achieve reliability and manage latency. This tool layer can be conceptually decomposed into several categories: information retrieval tools (e.g., web search, database queries), environment interaction tools (e.g., code execution engines, robotic actuators), and model-mediated tools (e.g., vision models). The architectural challenge here is—consistent with the trade-offs in topology selection—that of abstracting these heterogeneous functions into a uniform, queryable interface. A "tool registry"—analogous to device drivers in an operating system—allows agents to discover, evaluate, and invoke available functions dynamically. This design pattern is central to scalable enterprise architectures, where task and data planners break down user queries and route them to appropriate tools and data sources through a central registry [38]. The granularity of tool abstraction is critical. For instance, a "code interpreter" tool as a high-level service abstracts the execution environment, while the "unit test runner" provides a verification service. In tasks like code generation, such modular abstraction is essential for automating verification and refinement loops, thereby mitigating the risk of asserting hallucinated output as correct.

Building upon this tool integration, the design of how these external functionalities are structured and exposed must complement the chosen interaction topology. Rather than a monolithic entity, a multi-agent system interacting with the external world should be viewed as a federation of specialized modules—each responsible for a distinct capability. Modularity is not merely a software engineering practice but a cognitive architecture, defining the collaboration boundaries between different components. For example, a framework can be built on a hierarchical and self-organizing architecture where a manager agent dynamically generates subtasks and a group of worker agents directly address them without the need for new primitives [15]. This allows a more autonomous system that can expand or collapse to meet task constraints and exert control flows [39]. The external environment shapes the design space for agent-based system organization—the choice of topology—centralized, decentralized, hierarchical, or flat—serves a critical role in determining the flow of control, data, and influence [33, 40]. A centralized model, akin to the master-slave architecture in multi-agent deep reinforcement learning [41], where a lead agent possesses the global state and decomposes tasks, is straightforward to implement but notoriously introduces a central bottleneck and a single point of failure—a direct echo of the architectural tension described earlier. Conversely, a decentralized peer-to-peer architecture, where self-organized agents share information and adapt their communication pathways, directly addresses resilience issues by bypassing the centralized controller but demands sophisticated communication protocols and may introduce the corresponding overhead [42]. The relative strength of the external layer thus hinges on the chosen topology, which in turn determines the integration requirements for coherent operation.

At the outermost boundary, the concept of extension into comprehensive environments, including non-LLM components, marks the next frontier. The operational infrastructure must be capable of integrating external components that may include ML models, cognitive tools, or task-specific deterministic functions. A key design principle in this regard is the use of a centralized high-level "planner" that can leverage separate, purpose-built components to execute complex or computationally expensive tasks [38]. Such a system can directly bridge the LLM's planning capability with the existing system, thereby elevating the LLM to be the core operating unit—the "agentic kernel"—within a larger multi-agent software ecosystem [37]. An OS-inspired design that differentiates between the system kernel (LLM/Planner) and the application layer—the agents—allows for standardization and the efficient management of the entire multi-agent society, bringing the management of the system into the foreground [43]. This layer, therefore, becomes the structural bridge: it translates the systems design decisions made at the internal architecture global state, not only a selection but also the set of constraints for external integration.

In conclusion, as agents move toward actual deployment, they must bridge the gap between the abstract organizational schemes described in previous subsection and the future of the unstructured applications. The system software architecture is as crucial as the agent topology itself. The design is the framework, the application store analogy helps—the actuals for tools as the peripheral and the tool registry as the driver; therefore, the principle of abstraction and definition of *contracts*—between the centralized planner and the exposed tool—become the foundational layer. The externally-oriented architecture must be as carefully designed as the internal topology, and the choice of where to draw the boundaries is a design decision as consequential as the topology choice. The design of the external infrastructure is not merely a repository of functionalities but an implicit extension of the system's organizational principles, allowing for modularity, reliability, and composability in the broader open fabric of intelligent systems.

3 Communication, Interaction, and Coordination Mechanisms

3.1 Communication Protocols and Information Exchange Paradigms

The communication layer constitutes the foundational substrate upon which collective intelligence in LLM-based multi-agent systems is built. The design of this layer—spanning from message syntax and protocol structure to the semantic density of exchanged information—fundamentally determines the fidelity, efficiency, and expressiveness of group reasoning. This subsection analyzes the principal communication paradigms, their representational trade-offs, and the emerging control-flow considerations that shape their use in practice.

The predominant paradigm is free-form natural language dialogue, in which agents exchange textual messages that encode their reasoning, proposals, and critiques. This approach leverages the full linguistic and world knowledge of LLMs, enabling flexible problem-solving where agents can state assumptions, discuss alternatives, and arrive at nuanced conclusions in a human-like manner. Frameworks exemplifying this exchange include multi-agent debate and multi-model round-table conferences [10, 11], and it serves as the backbone for a broad family of collaborative agents [44, 13]. However, the unconstrained nature of this protocol introduces significant risks. Within closed-loop systems, errors can propagate across agents in a "cascading hallucination" effect, where increasingly fabricated statements are amplified through role-playing or debate as if they were received truth, degrading the quality of downstream output. This vulnerability is magnified when messages are over-communicated without rigorous grounding, leading to redundancy or persuasion-driven groupthink rather than information gain [21]. Furthermore, modern multi-platform architectures show that the ambiguity of pure natural language can be actively harmful; when the interaction is not governed by a structured contract, agents often fail at the communication level itself, suffering from breakdowns in instruction handling, interoperability, assumption management, distributed execution, grounding, and optimization [45].

To counter the ambiguity of raw text and to provide a more reliable substrate, the field has moved towards more structured symbolic protocols. These are characterized by explicit procedural rules or machine-readable message schemas that define communication operations within a group or workflow. Standardized framework architectures often exist between natural language and formal languages, imposing rigid data-format contracts or using tool-level calls and keywords for instructions while retaining flexibility [13, 46]. Adopting classical agent communication languages was another early effort, before LLMs became popular. These protocols are not incompatible; yet they have been largely eclipsed within the "prompting" paradigm because LLMs still operate best end-to-end as semantic parsers. However, the push for verifiability and control is strong, with designs like "Formal-LLM" [17] showing high integration of natural language planning and formal automata. Such hybrid systems allow human users to express control constraints; by integrating precise control flow with high-level language reasoning, these systems impose local compliant behavior on the agents and make the planning process tractable and auditable.

A more radical, and highly promising, paradigm is embedding-based communication, often termed "ciphered" or continuous communication. Instead of transmitting discrete linguistic tokens, high-dimensional continuous embeddings—from the LLM's internal representation—are directly transferred between agents. This act of transmitting "belief states" provides a much richer and denser format of information relative to discrete token generation, preserving a smoother support of the model's representation and avoiding the information bottleneck at token-sampling. Consequently, utterances of this form drastically reduce degeneracy from cascading deviation-prone textual propagation. The core insight here is that, by relieving the semantic item of the need for explicit tokenization, the communicative channel directly broadcasts a compressed view of the "internal state", thus enabling a more effective collective problem-solving process [47] and reducing message distortion for task-oriented coordination [31]. This is also especially important in mixed-modality settings, where the perception of the environment is non-textual [48] and the pure-textual routing fails both under large-scale heterogeneity and under missing ground-truth syntax [45].

These two paradigms (text and embedding) represent fundamentally different philosophies on how much internal cognitive format to commit to the environment. Natural-language protocols emphasize eliciting tacit reasoning from LLMs with other agents, which is expressive but computationally expensive and suffers from friction (through tokenization, serialization, and polling). Across separate agents, these losses can compound, making the system irrational in scale and consistency. The structured, symbolic approach trades the openness of communication for robustness and format compliance, and is required for deployment in high-reliability or regulatory environments where interaction logs must be auditable and predictable. In contrast, embedding-based side channels maximize "bandwidth" between agents, but this comes at a very real cost: and are continuous and opaque to inspection, impeding easy interpretability and context extraction. The choice among patient parcia—and their possible hybridizations—must be a trade-off between expressiveness and fidelity, between transparency and system scale.

The final dimension of a communication architecture is its placement within the overall control loop of the system. An agent architecture with multiple separate phases of behavior—such as complex planning, acting, and reflecting—must be explicitly designed as a **control flow**. Recent work has suggested that the synergy between multi-agent communication and reinforcement learning can improve coordination, and that communication protocols themselves can be adapted dynamically to environmental feedback [49]. The most compelling direction in this field lies in unifying these paradigms: primarily emitting continuous embeddings for maximum fidelity early in a long-horizon deductive task, and then constraining to a structured natural-language summary of the final consolidated plan for the human user (or coordinator) and relying on bounded syntactic output for readability [25]. This hybrid approach offers the best of both worlds—the efficiency and depth of searchable representations, amplified by the transparency of a controllable symbolic layer. Future research should push on these dynamics: not only a communication protocol as a fixed physical interface, but as an adaptive **strategy**, with resource types that are dynamically switched based on the uncertainty of the agent platform and the complexity of the subtask. In this view, core abstraction shifts from the format of how agents message to the functional role of what information flows with what fidelity.

3.2 Debate, Negotiation, and Consensus-Building Mechanisms

The resolution of conflicting viewpoints and the synthesis of distributed knowledge into a unified output constitute a defining challenge in LLM-based multi-agent systems. Building directly upon the communication substrate and organizational topologies discussed previously, these mechanisms must navigate divergent opinions, mitigate individual model fallacies, and produce outcomes that are more reliable than those of any single agent. This subsection examines the principal interaction paradigms—debate, negotiation, and voting—that govern these deliberative processes, detailing their workflows while critically analyzing the conditions under which they yield superior collective judgments. Just as the choice of communication protocol determines the fidelity of information exchange and the system's topology constrains its flow, the selection of a consensus mechanism sets the manner by which the collective reconciles the inputs it has received and produces its final decisions.

Multi-round debate represents the predominant paradigm for refining answers and suppressing hallucinations. The *ChatEval* framework employs a team of evaluator agents that engage in a collaborative discussion, leveraging the different capacities of each member to handle evaluation tasks like an actual human panel [50]. More sophisticated variants structure this exchange via role assignment: a proposer agent presents an initial solution, while adversary agents challenge its validity across multiple rounds, often converging on a more accurate consensus [21]. This dynamic in multi-agent debate has shown to improve self-consistency and correct errors that would otherwise persist in naive reasoning chains, an approach widely adopted in agent frameworks which configure dynamic debate settings [51]. A critical insight from this literature is that the efficacy of debate is not solely based on the size of the debate team, but on the balance between healthy disagreement and conformity; in settings with role-playing and predefined personality traits (e.g., overconfidence), groups tend to converge prematurely, meaning that controlling personality and prompting style is as important as the debate protocol itself [21]. This sensitivity reinforces the earlier observation on the expressiveness of free-form natural language: while debate leverages the full semantic range of language to interrogate viewpoints, its unconstrained nature also carries the risk of cascading hallucinations and persuasion-driven groupthink, requiring the judicious control of the size and autonomy of the debate.

When agents have divergent objectives, or when they possess private information, the interaction must be retooled into **negotiation**. Here, agent interactions align more closely to game-theoretic reasoning, relevant to bargaining, resource allocation, and the resolution of social dilemmas [52]. Notably, negotiation goes beyond pre-defined, templated exchanges, with the trajectory of communication shifting from information exchange to strategy and with agents developing increasingly elaborate strategic behaviors. This includes the observable emergence of deceptive but competitive strategies—such as pretending to be in distress—to achieve better payoffs [53]. This dynamic behavior requires agents to engage in theory-of-mind reasoning, where they infer partner incentives and adjust their own negotiation tactics based on belief states to negotiate more effectively [52]. These strategies hint at more effective negotiation on basic economic tasks (e.g., the ultimatum game) when compared to single model bargaining outcomes, enhancing our understanding of how more adaptive, environmental-goal-oriented communication protocols can lead to more nuanced negotiation in either cooperative or adversarial settings [54]. In this sense, negotiation naturally extends the decentralized and interactive topologies examined in the preceding section but adds a more explicit and strategic layer to the communicative strategy, aligning with the notion of communication as a dynamic, adaptive strategy rather than a static protocol.

The final component of this toolkit is **voting and aggregation**, a mechanism used when the goal is to synthesize the outputs by combining insights from diverse agents, especially when individual outputs have varying levels of confidence on the same final decision. In this framework, majority voting or ensemble weighting is used to select a final answer from the ensemble of individual outputs, a technique often employed in frameworks to improve reliability by assessing the contribution of particular members (or committees) of an agent team [22]. This approach aligns most directly with the structured protocols and decision-theoretic aggregation methods discussed earlier, offering a stable and well-defined output schema. More complex protocols employ hierarchical intermediaries, where a *judge* model or a more specialized meta-agent resolves disagreements and finalizes decisions—a dynamic role that is increasingly integrated into whole team structures. *Optimal team synthesis* provides strong evidence that a speedier, hasty aggregation strategy alone may not bring the highest collective intelligence, and that a tailored team dynamic structure, combined with a sequence of interactions (as enabled by the topological and communication choices above), is a more favorable alternative [22].

In examining the interplay among these mechanisms, the strengths of each become more clear across the context of the system's constraints. Adversarial debate can amplify the reasoning process to greater reliability but increases wall-clock time, token usage, and API calls—an unavoidable resource commitment for high-stakes reasoning, yet it is partially mitigated via more distributed scheduling and connectivity [55]. Negotiation offers a nuanced, strategy-aware agreement but requires careful balancing of objectives and higher cost through communication overhead. Voting provides a cost-effective consensus that requires less frequent token-heavy discussions, offering a crude but rapid distillation of the group's output; however, it ends with the depth of *why* the group decided a direction but not the reason, which within the context of a full deliberative framework places a heavier burden on the variability of the underlying inputs. To synthesize these diverse traits, a more deliberate division of work is often observed—an ability to leverage multi-agent capabilities adaptively reduces token calls required for tasks without sacrificing output quality, an efficiency that becomes increasingly critical as the architecture models grow (scaling topology) and communication volumes increase. Future developments will therefore not only focus on designing even richer deliberative protocols but also on **meta-cognitive strategies** that intelligently route the choice of debate, negotiation, or voting based on task characteristics, the estimated diversity of the agent pool, and the relevant performance metrics, resulting in an adaptive design space that is increasingly recognized as the central unifier of these paradigms.

3.3 Structural Topologies and Dynamic Organizational Adaptation

The organizational topology of an LLM-based multi-agent system constitutes the structural backbone that determines information flow, control authority, and the macro-level behavioral repertoire of the collective. These architectures occupy a spectrum from rigid, hierarchical pyramids to dynamic, self-organizing networks. The selection and adaptation of these topologies is not merely an engineering choice; it is a fundamental determinant of the system's capacity for collective intelligence, its resilience to failure, and its operational efficiency in complex environments.

At one extreme, hierarchical and centralized coordination structures are widely adopted in frameworks designed for complex, long-horizon tasks. In these layered top-down models [15], a central planner or "team leader" agent maintains a global view, decomposes a high-level objective into sub-tasks, and orchestrates a network of expert agents. This design ensures tight coupling between tasks and resources, enabling deterministic control and coherent strategy alignment from primary directives down to individual actions [56]. However, the concentration of authority introduces a critical bottleneck. The central coordinator easily becomes a single point of failure and a token and reasoning bottleneck, limiting scalability because all communication and planning flows through a single, finite input context [35]. Consequently, while centralized systems support high-level of coordination, they often struggle to be efficient in large-scale, open-ended environments that demand autonomous adaptation and localized decision-making.

In contrast, decentralized and flat organizational models embrace peer-to-peer interaction without a central authority. In these systems, agents share information directly, self-organize into teams or coalitions, and act independently to achieve global coherence. This architecture excels at navigating unpredictable environments, demonstrating high scalability and robust fault tolerance, as no single agent's failure can cripple the entire system [33]. To achieve this, such systems require emergent coordination mechanisms to form dynamic leadership or to generate new agents on the fly according to task complexity [18]. For instance, highly decentralized, bottom-up organizations can make better decentralized decisions based on local information [33] and decompose complex, long-horizon tasks into manageable sub-tasks [15]. This autonomy, however, comes at the cost of more complex coordination to reach team consensus [33].

Increasingly, the field recognizes that a single static architecture is insufficient for varied problems, and it is moving toward modular, reconfigurable systems. Instead of fixed hierarchies, many modern systems adopt a flexible philosophy where the communication path, task-flow structure, and even the team composition itself are determined adaptively during execution. Taking this a step further, these systems may employ a hierarchical, auto-organizing mechanism, which ensures centralized planning and decentralized execution for dynamic group adjustment, where agents can change their roles and collaborative patterns on-the-fly to adapt to the evolving states of the task [15]. These dynamic models are not limited to system-level behaviors; they also bridge the human-agent gap. In mixed human-agent settings, such topologies can be assembled dynamically given the current demands, reflecting real-world information asymmetry where humans and agents collaborate, and with open channels of communication that adapt to the task [57].

The organizational and architectural topology performs an important trade-off between control, resilience and scalability. Centralized systems offer high controllability and coherence but suffer indefinitely in high agent count environments, whereas decentralized, self-organizing structures improve scalability and robustness but increase coordination costs. The move towards dynamic and self-organizing topologies—shaped by evolving task requirements and environmental conditions—is (and will) define the future potential and ability of large language models in complex collective decision-making.

3.4 Deliberative Group Dynamics and Collective Emergent Reasoning

Deliberative group dynamics in LLM-based multi-agent systems represent a fundamental departure from monolithic reasoning paradigms, operationalizing Minsky's vision of a "society of mind" where distributed cognitive modules collaboratively solve problems that exceed the capacity of any single component [23]. Building directly upon the architectural topologies outlined previously, the deliberative processes described here illustrate how information flows and control authority are operationalized through iterative cognitive exchange, transforming static structural choices into dynamic collective reasoning. This emergent property arises not from individual agent intelligence but from the structured interplay of proposing, critiquing, and refining solutions across multiple agents—a process that mirrors how distinct functional areas of a biological brain cooperate. Research demonstrates that this interaction pattern prevents the "Degeneration of Thought" trap that plagues single-agent frameworks, where reasoning collapses into repetitive or circular logical structures [58]. By distributing cognitive labor and enabling iterative refinement through deliberative loops, the system collectively explores a far broader solution space than any agent could access independently, leveraging the adaptive, reconfigurable communication channels enabled by the flexible topologies discussed in the previous section.

The emergent collective intelligence observed in these systems manifests through cascading processes where the output of one agent acts as an informational substrate for the deliberations of another [33]. This reciprocal stimulation generates a positive feedback loop for coordination quality, as proposed in theoretical architectures for organizing LLM societies. The deliberation can be formally understood as a distributed optimization process: each agent performs local rational optimization and exchanges it through a communication mechanism. Successive iterations of refinement result in improved global solutions, achieving equilibrium or consensus points inaccessible to isolated reasoning. This resembles distributed estimation or control where interactions among agents result in effective synchronization behavior, albeit in the domain of symbolic meaning and logical inference through language generation processes. As such, deliberation becomes the operational layer that brings the topological diversity—whether centralized, decentralized, or dynamic—into the cognitive life, bridging the structural and the functional.

The macro-level dynamics of agent societies exhibit profound emergent properties. Empirical phenomena demonstrate how influence diffuses and governance structures arise from a set of autonomous agents, resembling the self-organization of social intelligence mechanisms observed in real societies [33]. In such social simulations, agents form a "wisdom of partisan crowds," where the cognitive diversity among specialized agents is a prerequisite for richer deliberation. This analysis suggests a complex interplay between the construction of the agent team as a social grouping and the canonical roles of a formal organization, human-in-the-loop chats, and individual-level adaptation. From a structural perspective, the conditions for emergence correlate with the presence of non-obstructive collaboration, allowing asynchronous task execution [42], and the implementation of an auto-organizing communication mechanism [15]—consistent with the reconfigurable and dynamic role of topologies detailed previously.

However, this collective emergent phenomenon introduces critical systemic vulnerabilities that demand rigorous attention. Unconstrained communication pathways—deliberately fostered for open-ended exploration—also enable the rapid propagation of false statements or harmful states, the leading to a "false consensus" problem where the group collectively reaches an incorrect conclusion with high confidence [58]. This systemic risk is intrinsic to the dynamic nature of emergent interaction that maximizes information flow to the collective. Designing robust group deliberation architectures therefore requires developing meta-level governance structures that balance deliberation the potential gain of emergent, communication-driven exploration against the negative effects of premature consensus. Future research should address this tension between the emergent, self-organizing forces central to the current paradigm and the necessary constraints that prevent systemic risks. This includes integrating more principled consensus mechanisms that preserve semantic correctness and enable cooperative recovery [59] while retaining the advantages of an evolving division of cognitive labor. Ultimately, the governance challenges introduced here define the necessary safeguards that will be explored in the following discussion of reliability and error resilience in multi-agent deliberation systems.

4 Collective Planning, Reasoning, and Learning in Multi-Agent Systems

4.1 Hierarchical Task Decomposition and Resource Allocation

The transformation of high-level objectives into executable multi-agent workflows constitutes the foundational layer of collective problem-solving. At the heart of this process lies a two-fold challenge: the decomposition of an abstract goal into a structured set of interdependent sub-tasks, and the subsequent allocation of these sub-tasks to specialized agents in a manner that respects their unique capabilities, communication constraints, and the finite computational resources of the system. Unlike classical multi-agent planning which often relies on predefined state-action spaces and explicit coordination protocols [2, 1], LLM-based systems exhibit a unique form of **semantic decomposition**. These systems leverage the general world knowledge embedded in the LLMs to perform "planning in natural language." This involves parsing a high-level directive into a hierarchical task graph—not merely a linear sequence—which explicitly encodes dependencies and allows for the parallel, multi-level orchestration from abstract directives down to concrete executable actions. This semantic structuring of a problem space is what separates LLM planners from classical symbolic planners, which are brittle and cannot gracefully handle the ambiguity of complex goals [60, 25].

Given a well-formed task structure, the system must solve the allocation problem—mapping agents to sub-tasks based on **capability modeling**. Early models such as *Role-playing* rely on static role assignments using hand-crafted game profiles [60, 16]. Emergent work, however, has shifted towards dynamic **coalition formation**. Systems like *Dynamic LLM-Agent Network (DyLAN)* actively assign agents to tasks based on an *inference-time* Agent Importance Score, an unsupervised metric that measures each agent's contribution to a sub-problem. Similarly, *Self-Organized Agents* (SoA) automatically multiply the number of agents based on problem complexity, thereby ensuring that the cognitive load of the entire problem does not collapse into a single utility's context window [42, 27]. A striking contrast between classical MARL role-learning [61, 62] and contemporary LLM-based systems is the mechanism of specialization. While roles are emergent clusters inferred from raw action-effects in policies, LLM-based coalition formation is driven by task relevance and semantic profiles. Consequently, task decomposition and role assignment share a tight coupling—as the semantics of planning into sub-problems evolve, the capabilities of the agents assigned to them must adapt accordingly.

However, real-world tasks are often **non-stationary**, and the environments that teams operate in—whether in embodied simulation or physical robotics—are prone to unpredictable disturbances. Adaptive planning, through Re-planning and re-allocation under dynamic constraints, is therefore an essential component. Frameworks like *ReAd* (RL-driven Advantage Feedback) treat the planner as a continuous optimizer. The process is a cooperative loop: an LLM planner proposes an action that maximizes a sequential advantage function in a Multi-Agent Reinforcement Learning loop, with undesirable actions corrected quickly [46]. Collective planning is thus not merely a static *pre-hoc* phase, but an iterative and grounded process that continuously seeks *online* strategy guidance and feedback.

Finally, the allocation problem has resource-based objectives aimed at **balancing multiple objectives**. The finite context window is the primary bottleneck for LLM agents, as their working memory cannot integrate a long interaction history without context overflows. Therefore, distributed working memory is necessary to support the "division of cognitive labor"—not only as a way to implement expert specialized roles, but also as a volatile "storage" mechanism that persists across agents. The design of the division-of-labor strategy is broad issues of latency and distributed token budgets between agents, requiring careful orchestration between competing objectives [63, 64].

Emerging trends point toward a move from designer-specified static workflows to self-organizing teams that are not frozen but evolve in response to the problem. One crucial next step is using *self-evolving agents* to realize long-horizon autonomy. The design of orchestrators that integrate human-in-the-loop feedback and environment-awareness also hints that dynamic re-planning is becoming the norm, not the exception. The governance and allocation of resources are thus gradually shifting from a centralized top-down system to an emergent *combination-pull* system, in which tasks are themselves a living directory of capabilities, to be bound and re-bound as circumstances dictate. As in many traditional social processes, this shift would give rise to the meta-level question of understanding the precise rules that yield optimal outcomes—an area warranting systematic study and evaluation to capture its full adaptive value.

4.2 Shared Deliberation and Strategic Reasoning in Multi-Agent Teams

Collective deliberation in LLM-based multi-agent systems transcends mere information exchange by establishing structured cognitive processes through which groups generate, challenge, and refine solutions, thereby engaging in distributed reasoning that often surpasses the capability of any individual model. The dominant paradigm explores how adversarial debate, structured verification, and the perspective interplay of specialized roles instantiate a "society of minds" [8]. If the previous discussion centered on how tasks are decomposed and agents are assigned, this subsection shifts focus to what happens *within* and *across* those agent teams once collaboration begins: how they collectively reason, contest, and converge. The foundational mechanism for this distributed cognition is multi-agent debate, first widely conceptualized by ChatEval, which posits that a team of LLMs engaged in structured discussion can produce evaluative judgments that are more reliable than a single agent-webs assessment [50]. Similar to the philosophical insights of Minsky, this approach treats reasoning not as a monolithic operation, but as an emergent group dynamic arising from a multi-perspective, iterative exchange [8]. Debate-centric approaches operate on the principle of adversarial verification, where distinct agents are assigned different roles (e.g., proposer and critic) with a shared context, creating a mechanism of collective self-correction that reduces individual model whims and mitigates cascading fabrications [50, 21]. Notably, this reliance on adversarial roles echoes the capability-driven allocation discussed earlier, but shifts the goal from task efficiency to epistemic robustness—validating that the decomposition of cognitive labor serves not only throughput, but also the integrity of the collective output.

Building upon the broad debate concept, research has expanded to model richer, multi-path search and search-like processes that introduce temporal dynamics and diversify benchmarking independence [8, 51]. A logical progression beyond linear debate involves concurrently exploring multiple solution branches—Tree-of-Thought and graph-perspective models—where distinct teams evaluate the rationality of specific intermediate steps and intelligently prune faulty paths to compose a final, verified answer [22]. These methods treat reasoning not as a decision pathway issued from a single model, but as a team-based search problem. Dynamic LLM-agent networks combine this debate with a centralized observer that adaptively modifies the team structure to amplify the most effective ideas, challenging the assumption that static architectures are broadly applicable—optimizing the assembly of collaborative agents leads to superior outcomes [22]. This extends to frameworks like Cross-Team Collaboration, where any team decision is communicated and evaluated by all other collaborating experts, thereby widening the concept of solution “quality” beyond a single metric [14]. The orchestration of such deliberation is, however, inherently constrained by the budgets and latency concerns taken up below; multi-branch exploration and recursive evaluation often incur overhead that must be continuously justified by gains in output reliability.

The advantages of these deliberation mechanisms extend beyond textual outputs to surface unique strategic and interpretive reasoning. A growing body of work examines the emergence of beliefs and behaviors in social environments that stretch far beyond conventional game maps [65, 54]. At the most generative extreme, the implementation of social games involving hidden roles, collaboration, and deception, such as the Avalon game, requires LLM agents to infer the mental states—beliefs, intentions, and knowledge—of others, evidencing a kind of *theory of mind* and strategic adjustments that inform task-oriented coordination [52, 65]. This alignment, however, is nuanced: in social psychology, agents sometimes exhibit deviations from formal logical behaviors, resembling human limitations and biases [53, 54]. An interesting phenomenon emerges where LLMs learn to strategically deviate from the “optimal” policy by bluffing to achieve a more favorable outcome in a simulated negotiation, which is evidence of emergent strategic communication because complex situations are inherently ambiguous [52, 54]. Such behaviors reinforce the idea that dynamic role and strategy modulation, already present in the planning layer, becomes equally essential during active deliberation.

Despite these apparent capabilities, the fundamental limitations of cooperative LLM-based approaches remain, especially regarding long-horizon planning and hallucination about task state [52]. A promising future direction is the design of organizations that adapt and optimize intricate social structures and communication protocols—perhaps by discovering enhanced organizational prompts that reduce communication overhead and improve team efficiency in multi-agent settings [30]. With this, the realization of role-based workflows using adaptive frameworks [51] will allow scaling of these emergent strategic reasoning abilities to more complex, socially meaningful tasks. The situated value of LLM-based multi-agent systems will be directly measured by their capacity to engage with the dynamic coordination of societies of intelligences—from social deduction to real-world collaborative design—ultimately feeding into the orchestration-level concerns of budget, adaptability, and human oversight discussed next [22].

4.3 Capabilities, Constraints, and Design Considerations for Orchestrators

The orchestration of LLM-based multi-agent systems embodies a fundamental tension: the pursuit of collective intelligence through sophisticated planning, reasoning, and learning mechanisms must be continuously reconciled with the hard constraints of finite computational resources, cognitive limits, and real-world deployment exigencies. The effectiveness of any orchestration strategy is thus not solely a matter of maximizing task performance, but of achieving a delicate equilibrium across a multi-faceted landscape of trade-offs.

Central to this equilibrium is the management of cognitive and latency overheads. Orchestration paradigms—from hierarchical decomposition to iterative debate and consensus-building—introduce substantial resource consumption in the form of API calls, token usage, and compute. The evaluative framework of *reasoning capacity* has been proposed to unify these constraints, defining the optimal problem-solving capacity achievable within computational budgets, token limits, and latency targets [36]. This perspective reveals that strategies such as multi-round debate [66] and exploration of reasoning paths, while yielding superior outputs, often incur an unsustainable cost for practical, time-sensitive applications. The challenge is amplified by the fixed context windows of LLMs, which constrain the collective's ability to synthesize information from numerous agents or long deliberation histories, necessitating sophisticated working memory management that prioritizes relevant information as subgoals are achieved [67].

The orchestration design is profoundly shaped by the operational environment. In scenarios guided by expert-designed workflows, the orchestrator must ground its planning in established process models to avoid the pitfalls of planning hallucinations and performance degradation due to its inherent lack of domain-specific knowledge. Conversely, in open-ended and unpredictable settings, a purely static structure becomes a liability. A comparative analysis reveals that formal organizations with predetermined roles reduce communication overhead in embodied tasks, but the most effective prompts for emergent teamwork are often those that the LLM agents themselves generate via a reflect-and-refine process to arrive at organizational prompts [30]. This suggests that the most robust orchestrators must support dynamic reconfiguration, creating fluid communication and task allocation mechanisms [15]. Without this adaptability, agents are confined to pre-defined roles, which forces them to rely on representations that can lead to task failures when encountering unseen or resource-constrained coordination scenarios [30, 15].

A critical design consideration is the integration of human oversight within the orchestration loop. Rather than serving as interactional appendages, human moderators play a pivotal role in guiding strategic learning and correcting multi-agent deliberations, thereby alleviating weaknesses in reasoning and enhancing overall system consistency [36]. This is especially vital in mixed-agent systems where human and LLM-driven agents collaborate, as their distinct strengths need to be embedded in the orchestration to solve complex tasks within safety boundaries and collaborative design templates [58, 68, 30]. The orchestration challenge is thus to design effective intervention and feedback pathways that preserve operational efficiency while ensuring high-quality outcomes.

Furthermore, the orchestration capability is fundamentally interwoven with the problem of knowledge generalization and inter-agent transfer. The manner in which agents are profiled and represented in a system directly influences interaction efficiency and grounded planning [25]. A successful orchestrator must therefore not only optimize for the immediate task, but also support the agents' ability to learn from experiences, update the collective knowledge, and transfer successful communication strategies to novel situations [13]. This points to a future where orchestration evolves from static task assignment to a meta-cognitive process—one that optimizes the system's structure, evaluates the adaptability and cooperation capability, and cultivates the collective's capacity for effective intelligence within its budget constraints [66].

5 Emergent Behaviors and Group Dynamics

5.1 Emergence of Societal Structures, Norms, and Conventions

One of the most compelling phenomena in LLM-based multi-agent systems is the spontaneous emergence of societal structures—leadership hierarchies, specialized roles, and implicit normative systems—that arise from decentralized interaction without explicit top-down programming. This self-organization is a hallmark of the paradigm shift from classical multi-agent frameworks that rely on predefined state-action spaces to contemporary language-driven agents that leverage flexible, natural language-mediated interactions [7, 8]. A central mechanism driving this emergence is role specialization. In open-ended task environments, agents frequently organically differentiate into coordinator, critic, and expert functions rather than executing preset prompts, a phenomenon explored in frameworks such as Dynamic LLM-Agent Networks, where agents are selected based on inference-time contribution and task conditions in a dynamic communication architecture [22]. Building on this, the notion of "collaborative generative agents" endows agents with consistent behavior patterns and task-solving abilities so they can automatically coordinate in task-oriented social contexts, revealing implicit role-taking behaviors [31].

This organic role differentiation often crystallizes into structural leadership and group hierarchies. Research shows that endowed organizational structures can further enable emergent leaders: embodied LLM agent teams demonstrate designated leadership dramatically improves team efficiency, while LLM agents spontaneously display leadership qualities and cooperative behaviors without being prompted [30]. However, the emergence of *de facto* leadership can also occur unaided by design. In networks, the variance in information centrality can create "hub" agents, whose influence and connectivity make them informal authorities that alter group dynamics and outputs [22, 18]. This is particularly visible in the emergence of roles from a framework where coordinating group of tasks involves emergent leadership; in Self-Organizing multi-Agent framework, agents automatically miniature and adjust groups according to problem complexity, enabling a type of distributed leadership to navigate open-ended multi-agent code generation and navigation tasks [23, 15].

Beyond role division, norms and implicit social contracts spontaneously consolidate within the group. Collective and ethical norms, as well as behavioral conventions, emerge through iterative interactions among agents, allowing ad hoc teams to coordinate without explicit pre-coordination. Studies investigating the collaboration mechanisms among LLM agents from a social psychology view have shown that multi-agent "societies" composed of LLM agents with different traits and thinking patterns (e.g., debate, reflection) display human-like social behaviors such as conformity and consensus reaching, thereby echoing foundational social psychology theories [21, 31]. This demonstrates that interaction of LLMs produce effect of "social norms" in ways not explicitly programmed. Further implicit social contracts are formed through consensus-based reasoning: ReConcile intensifies a structured round-table discussion resulting in refined group consensus via a confidence-weighted voting mechanism, and in that interplay, group deliberation is a collective meta-reasoning norm [11]. Similarly, multi-agent debate frameworks require a "judge" agent to manage the debate and achieve a final solution, establishing a norm of critical confrontation for consensus [10].

The phenomenon of *societal phase transitions* is especially intriguing yet under-researched, marked by the crossing of connectivity or consensus thresholds that turn a loose collection of agents into a cohesive problem-solving society. Research work on scaling LLM-based collaboration via multi-agent collaboration networks (MacNet) has demonstrated a collaborative scaling law: the normalized solution quality holds a logistic growth pattern as the number of collaborating agents scales. Crucially, they noticed a "collaborative emergence" that occurs much earlier than previously observed neural emergence; this points to a critical cluster size for the collective, after which emergent phenomena superlinearly enhance problem-solving [18]. However, this transition has a dual nature. While it enables superior group intelligence, the same phase transition can lead to costly echo-chambers if the group becomes too homogeneous or fragile if it transforms towards radical dispersion due to lack of diversity; indeed, debate frameworks show that multi-agent systems can also deviate from optimal reasoning depending on the balance of cooperation and competition [10, 21]. The result—whether the system crystallizes into a super-intelligent society or a homogenized faulty consensus—sharply depends on the diversity of the underlying models and communication topology [11, 18].

To advance this field, future research should move beyond the observation of these emergent phenomena towards understanding and steering them. As observed, [45] proposes a flexible and scalable platform for agent collaboration via an agent integration protocol and instant-message-like architecture, aimed at integrating heterogeneous agents in an Internet-like environment—a critical leap to building truly open, self-governing agent societies. The key is to aim for a system where emergent norms can be tracked, adjusted, and guided, providing a "nudge" for the community rather than a direct control mechanism. By synthesizing concepts from social psychology and organizational sociology in AI, we can, for instance, design mechanisms that promote emergent roles and shared blueprints in a way that keeps them aligned with the overarching goals of the collective.

5.2 Collaborative, Competitive, and Mixed-Incentive Dynamics

The interaction motifs that govern how LLM-based agents coexist range from fully cooperative endeavors to adversarial confrontations, with the majority of real-world tasks situated in a nuanced middle ground where incentives are partially aligned and partially conflicting. Unlike classical multi-agent systems where utility functions and action spaces are mathematically explicit, LLM-based agents negotiate these incentive regimes through natural language, making the emergent dynamics less predictable but significantly richer, often mirroring complex human socio-psychological behaviors [21]. In purely cooperative settings, we observe spontaneous information sharing and organic division of labor that transcend the initial role definitions—a direct consequence of the self-organization and role specialization discussed in the previous section. The AgentVerse framework [51] demonstrates that agents within a group dynamically adjust their composition and communication patterns to handle planning, discussion, and execution phases, leading to a greater-than-the-sum-of-its-parts outcome. Similarly, the study of embodied LLM agents in organized teams [30] reveals that imposing leadership structures reduces information redundancy, while the S^3 system [69] successfully reproduces population-level phenomena such as information propagation by endowing agents with the ability to perceive their informational environment and emulate interaction behaviors—tying these micro-level cooperative interactions to the macro-level social dynamics that will be examined in the next section.

Conversely, competitive dynamics exploit the adversarial capabilities of LLMs to challenge or refine outputs. The structured multi-agent debate paradigm, such as ChatEval's multi-agent referee team [50], demonstrates how a group of LLM agents engaged in discussion and debate yields richer and more nuanced and more reliable evaluations than any single-agent approach. These contest-like structures allow ideas to be propagated and challenged until a more stably refined solution emerges. This competition resembles the foundational configurations described in the previous section, where role specialization and social norms, in this case a norm of critical confrontation, arise from the interaction dynamics themselves.

However, the core of this subsection lies in mixed-incentive scenarios where individual rationality conflicts with collective goals, giving rise to emergent game-theoretic strategies that sit at the intersection of cooperation and conflict. The investigation into LLM-based agents in Avalon gameplay [65] provides a deep dive into these social dynamics, showing that agents in environments with hidden roles must balance civility, deception, and persuasion—phenomena that are not explicitly programmed but emerge from the interplay between language model capabilities and the environment. In social dilemmas, such as the Trust Game, LLM agents exhibit high behavioral alignment with human behavior, particularly for GPT-4, which demonstrates the potential of LLM agents to simulate a broad range of human trust behavior [53]. Similarly, testbeds in the Ultimatum Game [54] show the advantage of a multi-agent architecture over a single LLM. By simulating "greedy" and "fair" personality pairs, the multi-agent architecture achieved 88% accuracy in simulating human strategic behaviors, compared to 50% for a single LLM, specifically by capturing the tension between individual self-interest and social norms.

These findings corroborate the notion that emergent social equilibria, where agents navigate the delicate strategic collaboration, can be achieved only when multi-agent environments dynamically adapt [52]. This necessitates robust evaluation frameworks to measure the diverse capabilities that are required when moving from static to dynamic interaction settings [70].

Thus, as we progress, the focus will shift not only to understanding the design of the mixed-incentive structures and capabilities of LLM agents but also to evaluating the broader emergent outcomes of these systems. The micro-interaction motifs discussed here—ranging from subtle cooperation to adversarial competition—serve as the foundational building blocks for the macro-level societal dynamics and agent-based simulations that will be explored in the next section, providing the empirical and theoretical grounding needed to model complex social phenomena and collective intelligence.

5.3 Simulating and Modeling Human-like Social Phenomena

The advent of LLM-based multi-agent systems has inaugurated a new epoch in computational social science, transitioning agent-based modeling (ABM) from a simplistic, rule-driven paradigm to a platform capable of instantiating rich, cognitively plausible, and deeply human-like social actors [71]. Traditional ABMs often struggle to capture the bounded rationality, heterogeneity, and adaptive learning intrinsic to human behavior. LLMs, by contrast, provide an unprecedented substrate for encoding these complex traits, empowering synthetic agents with the ability to reason, plan, and communicate through natural language, thereby generating emergent macro-level phenomena that exhibit a remarkable degree of sociological authenticity [72, 66].

This enables researchers to probe social-psychological dynamics that were previously intractable, moving beyond simple behavioral heuristics. Crucially, LLM agents can embody cognitive biases, exhibit emotional responses, and engage in strategic misrepresentation, enabling the exploration of nuanced dynamics in classical game-theoretic scenarios. For example, the MAgIC benchmark deploys games such as Chameleon and Undercover alongside game theory scenarios (e.g., Cost Sharing, Multi-player Prisoner's Dilemma, Public Good) to create diverse testing environments that quantitatively evaluate the judgment, deception, coordination, and rationality of LLM-based agents [66]—revealing a capacity to mimic human-like strategic and communicative behaviors. The fidelity of these macro-level phenomena is largely predicated on the agents' ability to embody rich cognitive architectures, as frameworks like CGMI employ memory, reflection, and planning modules based on the ACT* model to facilitate realistic human interactions and social simulations [56].

The analytical reach of these systems is distinctly exemplified by their ability to replicate high-order social and spatial dynamics. LLM-driven simulations of urban and organizational environments, assessed through computational experiments, demonstrate the value of these agents in orchestrating and navigating complex social systems [71]. These systems do not merely reproduce static statistical patterns; their emergent interactions, often governed by the permeability of overlapping social networks, can precipitate the formation of opinion cascades and "information diffusion" dynamics, which are critical for understanding the "social physics" of online communities [73]. Furthermore, these agents show a capacity to navigate mutable leadership roles and organizational structures, dynamically auto-organizing into hierarchies to solve tasks in open-world environments [15]—which facilitates the study of self-organization.

The fidelity of these agent-based simulations and their capacity to harness collective intelligence for policy exploration ultimately pivot on the facilitation of interaction between the individual and the collective. The resultant macro-dynamics—such as human-agent teamwork at-scale—can even facilitate the co-existence and enhanced understanding of mixed human-agent interactions [74]. Yet, the utility of such emergent models rests on the ability to use the agent's authenticity to explore counterfactual histories and validate boundary conditions.

Despite this promise, significant challenges remain to realizing the potential of LLM-based social simulations. The absence of explicit guardrails in LLM-driven logical reasoning can be confounded by critical vulnerabilities, such as manipulated knowledge and miscommunication that promote dynamics like misinformation or bounded collaboration [75]. Critical future directions must therefore focus on anchoring these simulations in explicit cognition architectures to enhance control and interpretability, which itself should be brought into coherence with large-scale multi-agent challenges, such as grounding and mitigating hallucinations [76, 75]. As we move forward, the goal is to forge a platform that not only mirrors human mean-field dynamics but becomes a nascent programmable sociology in artificial intelligence frontier era—a precise, generative laboratory for understanding the very principles of cooperation, cultural evolution, and collective intelligence.

5.4 Conditions for Collective Intelligence and Systemic Vulnerabilities

The promise of LLM-based multi-agent systems rests on a central, yet often unexamined, premise: that structured interaction among multiple agents yields a superlinear increase in problem-solving capability. However, this emergent collective intelligence is not an unconditional property; it is contingent on specific architectural and dynamical conditions, and its manifestation is shadowed by systemic failure modes that degrade performance below that of any individual constituent. Building on the simulations of human-like social dynamics discussed previously, a critical analysis must therefore treat collective intelligence not as an inherent virtue of multi-agent systems, but as a phase in a complex system—one that is reached only under favorable conditions and remains susceptible to criticality.

The conditions for this super-linear capability can be loosely framed through the lens of scaling laws. Performance gains do not scale linearly with the number of agents; instead, a "logistic growth" pattern is often observed, where initial additions of agents yield significant performance increases that plateau as the system scales. This suggests a positive feedback loop between the self-refinement of individual agent outputs (e.g., through iterative self-reflection or debate) and the system's overall cohesion [23]. In this regime, the group acts as a verifier and an amplifier. Debate and formalized role specialization create a dynamical state where false premises are actively challenged, leading to robust system-level correctness. This is a foundational enabling condition: collective intelligence emerges when the system is explicitly designed to cross-examine individual agent outputs, preventing a single logical flaw from propagating unchecked through the pipeline. A further key enabling factor is the capacity for dense interconnectivity through per-agent topologies of heterogeneous experts, which enables more collaborative synergies [42, 77]. Organizational topology further modulates this process: hierarchical, modular designs can delegate functions without overburdening a central manager [33, 15], while decentralized flat structures achieve resilience through self-organization without central oversight [43].

In stark contrast, several systemic vulnerabilities can degrade this emergent intelligence, and these failure modes are not merely technical bugs but structural vulnerabilities that arise from the same interaction dynamics that enable collective gains. Chief among these is the problem of cascading errors. In a multi-agent architecture, a single erroneous output from one agent does not necessarily remain isolated. When agents are naively chained, a false premise tendered by a sub-agent can be adopted as ground truth by subsequent agents downstream, creating an **echo chamber** effect. This "error amplification" can cause the entire system to converge with high confidence on an incorrect answer. The very social dynamics that foster consensus in healthy systems become sources of coercive influence in these degraded states, leading to a form of "groupthink" that systematically suppresses dissenting voices and elevates conformity over correctness. The result can be a toxic escalation, where systematic logical errors are magnified across each generation of interaction rather than corrected [68].

This fragility is directly coupled to the system's robustness against a single erroneous seed, and is particularly visible in the social simulation environments discussed in the previous section, where one "bad actor" agent can rapidly cascade negative behavior if the communication topologies permit it to do so without effective verification bottlenecks [15, 23]. The vulnerability is thus proportional to the degree of interconnectedness unimpeded by strong verification loops: if communication is unregulated and uncritical, emergent dynamics favor information virality over information fidelity. This suggests that the conditions for collective intelligence (environmental variance, critical discussion) must be actively maintained, especially in the mixed-incentive scenarios highlighted in the preceding section. Conversely, when variance is suppressed for the sake of conformity, the system becomes brittle. Sheer network density alone is insufficient for information quality; for an agent to maintain a robust and accurate understanding of its problem state, we require a "**verifier-in-the-loop**" mechanism to resolve conflicts [39].

This analysis reveals that the crucial variables for collective intelligence are dynamical and interactional, not merely a function of agent counts. The field indicates that cooperative excellence emerges from a persistent and structured exchange of reasoned steps, combined with an absence of coercion among participating agents—conditions that prevent pathological conformity and the subsequent collapse of problem-solving autonomy. Consequenctally, robust management of heterogeneous agents to constrain the influence of a "bad idea" is not just a simple "safety feature"; it is the cornerstone of emergent intelligence itself. This defines a clear design imperative: materializing collective intelligence requires the ability to detect and rectify these systemic vulnerabilities during operation. However, this analysis also exposes a critical open frontier. We need a diagnostic framework that can reliably identify the parametric conditions that permit a "graceful degradation" of the system, rather than a sudden collapse, and we require principled metrics to predict when a multi-agent system is in a phase of superlinear growth as opposed to one of an echo chamber heading toward catastrophic failure. Such theoretical and quantitative tools will be essential for understanding how to prevent pathological convergence in multi-agent systems—and for charting the path toward the controllable, extensible framework that directly informs the broader considerations of vulnerability to information corruption and the need for systemic guardrails when scaling these systems to critical societal applications, which we turn to in the next section.

6 Evaluation Frameworks and Benchmark Design

6.1 Benchmark Suites and Testbeds for Multi-Agent Systems

The evaluation of LLM-based multi-agent systems presents a unique methodological challenge that distinguishes it from both single-agent assessment and classical multi-agent benchmarking. While single-agent benchmarks typically measure isolated reasoning or task completion capabilities [16, 28], and traditional multi-agent testbeds focus on low-level coordination in well-defined state spaces [1], contemporary LLM-based multi-agent benchmarks must contend with the open-ended, language-mediated, and emergent nature of agent interactions. Consequentially, the landscape of benchmark suites has diversified along two critical dimensions: domain specificity and the degree of environmental dynamism, with the latter capturing whether tasks are static problem instances or require continuous feedback-driven adaptation that elicits emergent coordination.

In domain-specific benchmarks, the emphasis is on stress-testing particular capabilities under ecologically valid conditions. For instance, the software engineering and code generation domain is extensively served by multi-agent collaboration frameworks that use code repositories as testbeds, where teams of programmer agents (each with designated sub-roles) must generate, review, and integrate code—as instantiated in benchmarks for self-organizing ultra-large-scale code generation and cross-team collaborative software development [23, 14], as well as in evaluation contexts from platforms like ChatDev [18]. These benchmarks often use pass-rate metrics (e.g., Pass@k) and agent-teaming architectural diversity as primary evaluative criteria.

Another domain, collaborative embodied interaction, underscores the necessity of multi-turn environmental feedback loops. For example, multi-robot collaboration benchmarks encompass a range of scenarios, from the physical coordination required to move obstacles to complex embodied navigation in simulated environments, as seen in work on embodied navigation systems with "auto-organizing" teams and the dialectic multi-robot collaboration framework benchmark [15, 20]. These testbeds specifically probe an agents' ability to reach consensus, allocate subtasks, and recover from state changes, a functionality absent in static code-generation tasks. Similarly, in the domain of interactive dialogue and debate, frameworks like Multi-Agent Debate [10] offer instances of commonsense machine translation and counter-intuitive arithmetic. In contrast, applications of a "society of minds" frameworks [11] provide benchmarks where multiple agents must negotiate and converge on an answer concerning divergent opinions, thereby testing consensus-building and self-consistency.

General-purpose evaluation suites are the paradigm of an alternative, emerging direction. As an illustration, AgentBench captures a multidimensional and evolving benchmark containing eight different environments—operating systems, databases, knowledge graphs, and card games—that evaluates the "LLM-as-Agent" abilities in a multi-turn, open-ended generation setting [16]. Similarly, frameworks like AgentGym test agent capabilities by having them act in diverse interactive environments, thus aiming to evaluate an agent's ability to learn and self-evolve based on feedback [28]. The distinction between "static" and "dynamic" is paramount here, and the most recent benchmarks are moving resolutely toward the latter, insisting on environmental fidelity as a prerequisite for evaluating emergent capabilities such as team formation, hierarchical planning, and common knowledge grounding, as emphasized in [18].

However, a critical gap persists: most benchmarks assess the final task outcome but may not explicitly evaluate the qualitative or quantitative performance of collaboration itself. The ability of an LLM-based agent team to organize into an ad-hoc hierarchy or divide cognitive labor may outperform a "sequential" baseline, but in a "single-agent-oracle" comparison, the impact of the topological choice on emergent intelligence might be conflated with other factors, such as the choice of agents or backbones [22, 63]. The development of benchmarks that include behavior-aware metrics, such as the observation of role-specific and task-centric states and their communications, is pressing. This requires shifting toward measuring agent interactions (e.g., token efficiency, information exchange) [63], rather than merely the aggregated product output.

Future work should explore the design of bespoke "dynamic-reconfiguration benchmark" where the optimal agent network topology itself is unknown, thereby probing a system's ability to self-organize and scale efficiently to potentially large populations in a high-fidelity simulation setting, echoing the ambitions of social-network simulation systems aiming to model citizen behavior in large social networks [69].

6.2 Performance Metrics and Efficiency Evaluation

Evaluating LLM-based multi-agent systems demands a methodological shift from conventional single-agent assessment. While task completion remains the primary success criterion, it is insufficient for characterizing the value proposition of multi-agent collaboration, which often introduces additional computational and economic costs. A truly comprehensive evaluation must therefore go beyond final-output metrics alone to incorporate two complementary dimensions: efficiency—quantified in terms of token consumption, latency, and communication overhead—and process-oriented benefits of collaboration, such as improved reasoning quality and robustness that may not be captured by final output accuracy alone. This dual focus is essential because the same collaborative architecture that yields emergent reasoning gains can also introduce non-trivial resource bottlenecks that, if unmeasured, obscure the practical viability of the system. For instance, research on dynamic agent team optimization demonstrates that while multi-agent inference incurs higher resource consumption than a single run, selective agent participation and early-stopping mechanisms can yield substantial accuracy improvements—13.0% on MATH and 13.3% on HumanEval relative to a single GPT-3.5-turbo execution—with reasonable computational overhead, underscoring the importance of reporting efficiency alongside effectiveness [22].

At the individual task level, the most straightforward metrics are exact-match success rates and task completion time, which provide an initial signal of system capability. Yet, modern multi-agent applications tackle long-horizon, knowledge-intensive tasks where binary success is often insufficient. The AgentBoard uses a fine-grained progress rate metric that captures incremental advancements toward a goal across multi-turn interactions, addressing the limitation of final-state-only evaluation in partially observable, dynamic environments [78]. Similarly, benchmark suites like Melting Pot evaluate continuous generalization and adaptation to novel social situations, a measure distinct from simple success rates in static tasks [79]. These process-oriented criteria allow evaluators to distinguish between a system that reaches a correct final answer through speculation versus one that progresses reliably through coherent intermediate steps, providing a more diagnostic view of agent capability.

Complementing task effectiveness is the quantitative assessment of resource utilization. Key indicators include token consumption, the frequency and serialized versus parallel nature of agent interactions, and the cognitive load imposed by context switching. The number of required interactions is particularly relevant in team-based reasoning; studies show that certain collaboration mechanisms—such as reflection-based debates—can achieve superior results with fewer API tokens, directly optimizing efficiency [21]. In a related vein, the orchestration architecture is an important efficiency parameter. The token budget and latency become the primary bottlenecks when moving from a monolithic single agent to a team of specialized agents, as each additional agent introduces message-passing and processing overhead. However, structured interaction paradigms in collaborative debate frameworks [50] show that overhead can be well-managed when the interaction topology is designed to converge, suggesting that latency metrics should be reported alongside a normalized complexity measure—e.g., per-turn reasoning depth against cumulative token usage.

The measurement of token cost and time is often complemented by economic scales. For many deployments, the financial cost of LLM calls is a first-order constraint, altering the trade-off landscape between performance and budget. As with distributed computational systems, strong scaling highlights the necessity of avoiding quadratic overhead from all-to-all communication patterns [80]. Consequently, when assessing multi-agent systems, the total cost of ownership—API calls, resource provisioning, latency—should be evaluated against task-level performance gains. For instance, the AgentScope framework illustrates that deployments actively optimize the overhead of repeated model inference and parsing, reinforcing the principle that any evaluation should include the total computation expenditure and its associated costs, not just accuracy [81].

A holistic evaluation framework also includes measuring the marginal value of collaboration itself, through what can be conceptualized as collaboration reward and orchestration quality. This analysis asks not just "how good is the final answer?" but "how much better is the answer because of how the agents collaborated?" Performance comparisons between single-agent and multi-agent designs, conducted under identical token budgets, illuminate the crossover point at which collaboration becomes beneficial relative to a monolithic system—a point often obscured when only the absolute final score is reported. Indeed, we should also note as multimodal agent surveys observe, different agent architectures represent a category variable that standard evaluation pipelines often miss [48]. The pivotal metric here is the marginal return on collaboration (the added value per unit of added computation), rather than the final score in isolation. To measure this, composite metrics are necessary; these track both task consistency: component goal completion and planning accuracy—and variance in the aggregated outcome. The AgentLite library explicitly targets this facilitation, providing a customizable toolbox for reasoning-strategy optimization in single- versus multi-agent roles [55].

Taken together, these efficiency, process, and orchestration-quality metrics must be integrated with the core effectiveness benchmarks that are already reported in the field. Rather than treating accuracy and cost as independent axes, the next generation of performance evaluation should adopt a multi-objective evaluation protocol that reports: (1) effectiveness scores under explicit and controlled resource budgets (e.g., a fixed token/latency envelope); (2) the marginal value of collaboration—computed as the performance delta between a multi-agent system and a single-agent baseline under equal constraints, thus disentangling collaboration gains from raw model capability; and (3) backward operationalized cost metrics that quantify the total computational expenditure per improvement unit. This framework, in addition to the robustness- and security-oriented verdicts described in the next section, provides a rigorous foundation for validating the practical and scientific worth of LLM-based multi-agent systems.

6.3 Safety, Robustness, and Failure Mode Analysis

The evaluation of LLM-based multi-agent systems must extend beyond task performance to interrogate the unique vulnerabilities introduced by their distributed, communicative, and emergent nature. Unlike monolithic models, multi-agent systems present a radically expanded attack surface where latent risks—such as hallucination propagation, adversarial manipulation, and the emergence of harmful social dynamics—are not simply the sum of individual agent failures but are compounded by the interactions themselves. Therefore, a robust evaluation framework must incorporate a taxonomy of stress-tests and verification protocols designed to expose systemic defects before deployment, focusing on (i) adversarial security testing of the collective, (ii) robustness to cascading errors and emergent group-think, and (iii) alignment audits within a safety envelope.

Adversarial security testing for multi-agent systems must go beyond single-agent jailbreaks to investigate attacks that exploit the communication fabric itself, including the spread of manipulated knowledge or the injection of malicious instructions through the disguise of a message from a peer. A significant risk is the "flooding" of manipulated knowledge, where a single compromised agent can covertly inject counterfactual or toxic information that propagates through the conversation network, corrupting the group's consensus and long-term shared memory, and persisting even within retrieval-augmented frameworks. Red-team simulation, where adversarial agents attempt explicit harmful outputs or backdoor system behavior, is a critical component of security testing. These evaluations must incorporate attacks such as role definition injection, where the opponent attempts to alter the core directives of an agent through natural language, or the deliberate malicious breakdown of an agent's persona and identity, thereby compromising the system's security and alignment [82, 76]. The threat model is further complicated by information asymmetry; with each agent holding private information about a user, a compromised agent becomes a major vector for leaking privileged data during inter-agent communication. Robust certifications must therefore verify not only the resilience of an individual agent but the security of the entire information flow, ensuring that a single flawed agent interaction cannot escalate into a system-wide breach [57].

In our collective cognitive dynamics, a critical failure mode is the establishment of echo chambers, herding, or the unauthorized spread of errors—what we might call a cascading error or a social conformity failure. This "hallucination avalanche" emerges when faulty outputs from one agent are amplified through a positive feedback loop of imitation, leading to a group-level overconfidence in a false collective state and the proliferation of errors. The architecture's very strength—the use of contextual information—can become a weakness, as each agent's observation of prior discussions risks creating a global homogeneity that suppresses independent critical thinking. The topology of the system itself contributes to the macro-level social dynamics, and the diffusion of influence within the network affects the speed of convergence to consensus and its accuracy, where homogeneous networks may accelerate consensus but fail to reject systemic error, while decentralized, diverse, and well-connected topologies are more robust to localized perturbations but can be degraded by specific network effects [73, 18]. To assess these failures, evaluation protocols must move beyond simple error rate to measure the *rate* and *spread* of a single, intentional false assumption as it propagates to the entire system. There's an important role for the “Theory of Mind” inference, as an unexpected adversarial or non-cooperative agent is not only a security vulnerability but also a test of the group’s ability to detect and deprioritize actors that are behaving in deviant ways [83]. Thus, robustness testing includes injecting a controlled “noisy” agent with corrupted decision logic to see if the system can either isolate the agent or engage in a high degree of counter-party deliberation to protect the group’s integrity and rationality.

Finally, safety evaluation must move away from a final assessment to a continuous verification protocol, an alignment audit. This involves testing the utility of the entire collective under a set of safety goals, ensuring not just that the final output is correct, but that the decision-making process and no single agent draws an unsafe action. Vulnerable points in the process of goal negotiation can create a situation of an "emergent misalignment," that is, a suboptimal compromise objective maximized by the group but adversarial to the original user intent. The overall resilience of the team can benefit from additional oversight, such as placing a specialized "guardian" agent in the network to monitor system status and verify factuality [76]. Such evaluations can be quantified with "safe action" metrics, where the evaluator defines an envelope of permitted commands and fails the group if it makes an unsafe command. Central to this auditing process is the ability to trace and infer the internal beliefs and intentions of agents to provide a transparent overview of collective decision-making [83]. As the community moves beyond centralized orchestrators, the certification of multi-agent systems hinges on their ability to perform well on ordinary tasks and to do so while navigating the deluge of potential systemic risks, proving that the collective is not fundamentally weakened by the presence of a few dynamic parts, but robust in the face of emergent failures.

6.4 Human-Centered, Collaborative and Believability Evaluation

The evaluation of LLM-based multi-agent systems has predominantly focused on task completion and efficiency metrics, yet a comprehensive assessment framework must extend beyond the artifact produced to interrogate the *process* through which human and machine agents collaborate. While the preceding discussion examined how these systems fail under adversarial pressure and systemic stress, a complete evaluation must also consider how they succeed and adapt in the presence of human partners—not merely in terms of functional output, but in terms of the quality of the socio-technical interactions themselves. Human-centered evaluation occupies a critical intersection: it must simultaneously verify that the system faithfully adheres to human behavioral norms, that its collaborative behaviors are perceived as natural and useful by human participants, and that the agents themselves achieve a degree of *believability*—the quality of being perceived as coherent, purposeful, and socially situated actors rather than brittle response generators. This dimension of assessment is fundamentally distinct from measuring task accuracy or stress resilience, as it shifts the object of study to the socio-technical interface where calibrated trust, interpretable intent, and role-appropriate behavior ultimately determine the viability of human-agent teams in real-world deployment.

The construct of *believability* has emerged as a central organizing principle for evaluating the human-fidelity of agent behavior in social simulations. Drawing on the "Turing test" tradition but moving markedly beyond simple imitation, several quantitative and qualitative frameworks are being advanced that decompose believability into measurable components—such as behavioral consistency, and the capacity for nuanced reasoning in self-interested negotiation. Rather than requiring perfect behavioral equivalence, these methodologies instead stress *consistency* and *robustness* to input perturbations, acknowledging that the goal is not literal human equivalence but *believable* simulation of human-like dynamics that can sustain meaningful collaboration. Furthermore, validation methods such as the Virtual Overlay Multi-Agent System (VOMAS) approach [84] offer a foundational blueprint: overlay agents monitor the base system’s social interactions against formal constraints that encode human expectations, turning the abstract quality of believability into an externally verifiable condition.

Remarkably, evaluations of human-agent team performance are re-orienting away from simplistic human preference studies towards a richer focus on human *perception* and *cognitive alignment*. Metrics such as "trust" are no longer treated as a unidimensional score but are recognized as a systemic and multi-faceted phenomenon contingent on the human's *mental models* of system capability, decision provenance, and the believability of the collective reasoning path taken. This interaction echoes the perspective that structured architectural blueprints—such as those proposed for foundation-model-based agents [85]—can serve as a substrate for human expectation-setting. When preliminary role assignments, functional boundaries, and the delegation of agency are architecturally defined, the human user can calibrate their mental model of the AI system, improving the measurability of its collaborative competence. Consequently, successful evaluation in a human-centric sense must necessarily probe the system's capability to *cooperate within teams*, which goes beyond task execution to incorporate transparent communication and auditable internal reasoning. This calls for a careful alignment of system autonomy with human oversight, a balance articulated in recent taxonomies of autonomy [40] that characterize degrees of human-in-the-loop and human-on-the-loop control.

Looking forward, continued progress hinges on systematically linking the internal functioning of a system—its architecture and reasoning—to its observable capacity as a *believable participant* within human-AI teams. A sustained evaluation of these behaviors requires a shift from static post-hoc measurements towards a dynamic, feedback-driven methodological toolbox. The field's future likely lies in the construction of collaborative benchmarks where, instead of evaluating the agent solely as an isolated actor, the key ground truth is the *fidelity of the joint human-AI task trajectory*. This comprises maintaining common ground, adhering to collaborative norms, and effectively demonstrating trust calibration; insights from established MAS engineering principles [86, 87] provide a starting point for such operationalization. Ultimately, the aim is to move beyond post-hoc scoring toward a compositional understanding of human-AI interaction as the foundation of evaluation, complementing the safety and security checks described in the previous discussion and setting the stage for the final evaluative dimensions proposed in this survey.

7 Challenges, Limitations, and Open Research Frontiers

7.1 Theoretical Underpinnings and Formal Verification

The current state of LLM-based multi-agent systems (LLMAS) presents a striking paradox: while empirical studies demonstrate remarkable emergent capabilities in problem-solving, reasoning, and social simulation [18, 27], the field lacks a principled mathematical foundation capable of explaining *why* these systems work, *when* they fail, and *how* their behavior can be formally guaranteed. The challenge is exacerbated by the unique nature of these architectures, which combine stochastic neural components with complex, often non-Markovian, interaction protocols mediated by natural language. Consequently, classical formal methods designed for deterministic or well-characterized stochastic systems, such as probabilistic model checking or standard Markov decision processes, are critically constrained in their applicability, as they cannot easily account for the open-ended, high-dimensional, and semantically rich state spaces that LLM agents generate and inhabit.

A principal theoretical obstacle lies in establishing formal models for emergent dynamics and convergence. Classical multi-agent systems benefit from well-defined rules of interaction [1], allowing for the application of fixed-point theorems or Lyapunov stability analysis to predict system convergence. However, LLM agents exhibit complex adaptive behaviors driven by in-context learning, making their collective trajectory a function of the entire interaction history and the specific prompts or protocols employed. For instance, the phenomenon of cascading hallucinations, where an erroneous output from one agent propagates and is amplified through its peers, underscores the urgent need to understand the information-theoretic conditions under which collective error is dampened rather than amplified. As the field moves toward more dynamic and self-organizing structures [15], static analyses become insufficient, demanding the development of constructive frameworks for *learning-based verification*. The challenge is not simply to verify a fixed configuration but to reason about the model's performance and robustness across the distribution of emergent interaction states.

A second pillar of the theoretical gap is the difficulty of conducting rigorous verification via model checking. Traditional techniques such as model checking rely on exhaustive exploration of the state space. The state space of an LLM-agent society is not discrete or finite; it encompasses the model's internal parameters, the entire context window of prompts and past communications, and a virtually unbounded set of possible action sequences. Initial forays into formal methods, such as combining PDDL-based classical planners with LLM heuristics [88] or injecting formal automata specifications to guide planning [17], have achieved some success on constrained tasks. However, these methods scale poorly to open-ended, generative environments. This raises the critical question of *abstract interpretation*: how to construct tractable abstractions of the agent system that preserve the essential logical, temporal, and epistemic properties of the interaction while discarding computationally intractable details. The analytical difficulty is compounded by the need to verify properties against meta-level mechanisms—such as the organizational topology [89] or the communication protocol—that fundamentally alter macro-behavior but are not visible in any single agent's internal state.

Beyond tractability lie the fundamental questions of decidability and theoretical computability: what can and cannot be achieved or guaranteed by any system comprised of interacting LLMs? Drawing on the classical work on agent communication protocols and the algebraic semantics of multi-agent interaction [6, 4], we can begin to ask formal questions. Can we establish limits on the power of collective reasoning—perhaps finding that *plurality and consensus-building mechanisms* cannot guarantee convergence to a global optimum in the presence of agents with non-comparable or partially-conflicting goals? The "impossibility of collective intelligence" is a plausible theorem in the context of LLM-based agents, especially when the utility functions of individuals are not aligned with a system-level objective. The very mechanisms that drive cooperation, such as debate and voting [11], can lead to "groupthink" and a false consensus [21], which acts as a mode of failure where a high-confidence incorrect answer is propagated. Formalizing these bounds means asking foundational questions about representation. We must move beyond evaluating the agent's isolated outputs to developing formal, testable criteria for the *joint* properties of its state. The digital "collectives" also necessitate a new perspective on deductive reasoning: unlike deterministic planners that can guarantee the attainment of a goal according to a pre-condition, LLM-agents operate in the space of *open-world assumptions*, where lack of information is pervasive and evaluating the effect of an action is a cognitive-linguistic act.

Finally, the lack of formal verification opens a dangerous intersection with robustness and safety. The individual agents that constitute an LLM team are themselves stochastic and potentially hallucinating, and the uncertainty of their collective behavior creates a massive surface for cascading errors. A formal framework must be developed to define, analyze, and prevent these failure modes, addressing the fundamental tension between individual agent autonomy and the safety of the collective. This demands a progress from a purely empirical, black-box assessment to a principled framework that can be used to certify the *probabilistic* end of irreversible harmful behavior. Building this will require a synthesis of distributed systems theory, epistemic logic, nonlinear dynamical systems, and computational social choice to form the theoretical 'science' of multi-agent LLM collaboration. The field must move beyond treating LLM-based systems as magic to understand them as engineering artifacts. This requires that the research discipline will be guided by appropriate methodologies and its progress particularly for generating a new scientific foundation of collective intelligence—a foundational question for the future of AI.

7.2 Scalability, Efficiency, and Computational Feasibility

Scalability remains the most consequential impediment to translating the promise of LLM-based multi-agent systems from laboratory demonstrations to real-world deployment. While the preceding analysis has highlighted the theoretical gaps in formalizing emergent dynamics and verification, these challenges are dramatically amplified by practical scaling constraints—indeed, the very complexity that defies formal analysis stems in large part from the combinatorial explosion of interactions that scalability forces upon us. The prevailing paradigm of fully-connected agent communication is dominated by an $O(n^2)$ combinatorial explosion in both interaction frequency and computational cost, rendering systems with more than a handful of participants practically intractable. As the number of agents grows, the token consumption of pairwise discussions, the latency of sequential coordination, and the financial overhead of repeated API calls scale multiplicatively, imposing an economic barrier that is frequently more binding than a purely algorithmic one. The challenge is further compounded by finite context windows, which introduce a cognitive bottleneck for individual agents within large collectives: once a central coordinator or a fully-connected peer must ingest messages from all other members, the informational content of the group quickly exceeds the model's maximum sequence length, forcing premature summarization, information loss, or complete system failure [8, 26]. Thus, a central tension emerges: the organizational structures most robust to individual voice failure are often those least scalable in computational cost [80].

Addressing the communication bottleneck has motivated a paradigm shift from all-to-all interaction patterns toward sparse, structured, and adaptive topologies. This shift is not merely an engineering convenience; it is a prerequisite for enabling the dynamic reconfiguration and continuous evolution discussed in the following section, as adaptive teams cannot be built upon a rigid foundation of quadratic communication costs. Hierarchical coordination and information compression—where a limited number of manager agents synthesize group state—directly mitigate the cost of dense communication by introducing hierarchical abstraction layers. This is observable in frameworks that employ manager levels or cross-team collaboration, where upper-level agents buffer and summarize the output of lower-level expert agents before passing the condensed state to a top-level planner [51, 14]. Such structural sparsity is a key paradigm for scalable coordination, drawing inspiration from small-world network topologies in classical multi-agent literature. This sparsity reduces the per-agent load while enabling multi-round, costly debate only within targeted sub-groups, interleaved with strategic summarized communication at the group level [21]. Further, the **principle of information frugality**—where agents query the shared state or other agents when a specific epistemic necessity arises, rather than executing a periodic all-to-all synchronization—has shown to produce asymptotic efficiency improvements, thereby allowing for a more measurable cost of coordination [90]. Such frugality aligns naturally with the learning and adaptation mechanisms described in the previous discussion, as it enables the system to allocate computational resources toward the information-theoretic bottlenecks that matter most. Concurrently, dynamic team construction methods that prune or add agents based on an *Agent Importance Score* during inference enable robust performance with a substantially lower wall-clock time and token footprint, echoing the efficiency of scale-efficient model selection [22].

The economic and memory viability of such system is a decisive factor. While systems may achieve up to 590 agents by relying on synchronous messaging (creating cost) and simplified interaction loops, platform engineering focused on message-passing, data serialization/deserialization formats, and actor-based concurrency is critical for enabling distributed deployment [81]. The cost of LLM inference is as significant as any computational cost, and the token consumption is magnified in multi-agent systems. Techniques for tool integration and compact data representation (e.g., concise tool instructions) and automatic graph-based prompt optimization [91, 89] are essential components of a cost-minimization strategy. Hosting millions of agents is a step beyond what current GPU-accelerated simulations can easily address, yet substituting heavy multi-agent LLM calls with lightweight, RL-based approximations—a hybrid design—can achieve qualitatively similar group behavior in complex tasks at a fraction of the cost of full large language model inference [79, 61]. Importantly, parallelism in current multi-agent inference (e.g., simultaneous independent agent generation) is able to yield low latency overhead compared to serial execution due to massive parallel GPU calls.

Beyond static limitations on inference cost, scalability is deeply intertwined with learning in the multi-agent reinforcement learning (MARL) regime, where failure to scale to large populations stems from the computational instability of modeling hidden states and the inability to target goals. To mitigate this, population-based meta-learning methods—which dynamically generate policies—are necessary to allow agents to generalize to unseen agents and environments as an alternative to full-retraining of large models [92] [93]. Structural regularizations, such as factorized value functions considering the actions of entity sub-groups (e.g., the Randomized Entity-wise Factorization) [94] also help scaling across tasks, achieving a dramatic improvement in learning. These approaches parallel the architecture focus of natural-language agent populations, where a small structural sparsity or hyper-network prior can induce coordinated behavior without centralized supervision and promote the expected group-level correlation.

Looking forward, the path to a genuinely large-scale ecosystem demands a multi-scale strategy. First, the adoption of agent operating systems—defined around data-driven scheduling and the principles of memory- and state-based organization—allows us to incorporate communications cost and latency directly into configuration control models—performing a multi-resource feasibility check is required for formal construction before execution—rather than relying on ad hoc protocols [95]. Second, to maintain affordable computation costs and ensure quality, the learning must go from emergent multi-agent interactions to *goal-gradient-based* learning across agents—leveraging efficient evaluators (fine-grained progress metrics) [78] and minimally redundant state representations to offer principled estimates of value. Finally, while scaling interaction often yields diminishing returns, it also enables negative emergent behaviors. Thus, prioritizing the *right kind* of sparse communication and using signals with real-time, adaptive capabilities to reduce or rearrange interactions based on a low-level statistical budget is a critical open challenge of the field in its pursuit of robustly scalable collective intelligence—one that builds upon the theoretical and architectural foundations laid here to explore the dynamic, learning-driven systems that follow.

7.3 Continuous Evolution, Learning, and Dynamic Reconfiguration

The prevailing paradigm in LLM-based multi-agent systems remains fundamentally static: agents are instantiated with fixed personas, communication topologies are established at design time, and organizational structures persist unchanged throughout task execution [8]. This rigidity stands in stark contrast to the dynamic, non-stationary environments these systems are increasingly deployed within, exposing a critical disparity between the architectural assumptions of current frameworks and the demands of real-world adaptation [27]. The challenge of continuous evolution, therefore, encompasses not merely the capacity for individual agents to update their knowledge, but the far more profound ability of the collective to reconfigure its own architecture, refine its coordination policies, and learn at the group level over extended horizons.

At the level of the individual agent, progress has been made through memory-driven architectures that permit in-context learning without parameter updates. Frameworks that integrate long-term experience memory with reinforcement learning signal mechanisms enable semi-parametric evolution, allowing agents to improve their decisions through performance feedback and experiential accumulation without costly fine-tuning [82]. The alignment of such memory modules with human cognitive structures—such as layered memory for expert agents in domains like financial trading—demonstrates tangible benefits, allowing the agent to adapt and refine specialized knowledge in response to volatile environments [96]. These approaches exemplify the shift from static "train-and-deploy" models to a paradigm of continuous dynamic evolution of member-agent policies. However, individual-level adaptation alone is insufficient; the coordination of decisions and abilities within a group introduces the necessity for synchronization of these learning processes across the collective.

The more scientifically compelling frontier, however, lies in the automatic generation, evolution, and dynamic reconfiguration of the agent team itself. Recent work has increasingly focused on systems that do not simply update their policies but expand their own structure through the automatic generation of new expert agents from a base framework, enabling the system to evolve in composition and size in response to complex task demands [30]. Such approaches are underpinned by evolutionary algorithms that create, modify, and select agents, and crucially, involve methods for cross-generation enhancement, including using self-reflection and LLM-as-a-judge techniques to optimize agent team topology without human intervention. This organizational learning extends to the process level. Methods like shared memory and ranking-based reflection allow the entire group to accumulate collective experience and evaluate its behavior to improve safety, effectively distributing knowledge across agents [32]. Similarly, dynamic frameworks that structure interactions as graphs with topological ordering show good scalability and facilitate the emergence of collective intelligence, including through the dynamic hierarchy and self-organizing communication mechanisms that allow agents to reorganize group structure for subtasks and *ad hoc* teaming [18, 15].

The capacity for learning must also be extended to the interaction level for macro-level coordination dynamics. In settings where multiple agents have distinct yet coordinated sub-goals, the emergence of intelligent reasoning from inter-agent communication, such as multi-agent debate and consensus structured via a global workspace, demonstrates that we can optimize collective thinking [97]. Such social dynamics rely not only on the individual agents' ability to execute subtasks, but also on the system's capability to orchestrate debate, reflection, and majority voting in a way that culminates in a system-level collaborative intelligence that surpasses the sum of individual model performances [66]. However, the trajectory of the field demands a movement from approaches that optimize coordination at a single level to an integrated approach where dynamic learning across individual and collective levels are intertwined. A key unresolved challenge is the creation of a system able to continuously showcase, in a way that optimizes its own self-organization. Addressing this will require a paradigm shift from the architecture able to act and co-evolve, in a self-organizing world, to a new focus on the creation of inherently open, generative and adaptable multi-agent systems capable of continuously exploring possible future collective behaviors.

7.4 Governance, Trust, and Socio-Technical System Design

As LLM-based multi-agent systems transition from laboratory prototypes to deployed socio-technical infrastructures, a triad of intertwined challenges emerges: governance, trust, and the design of human-machine collaborative environments. The governance of these systems extends beyond mere technical coordination into questions of accountability, compliance, and institutional oversight. Architectures that enable autonomous multi-agent operation must be complemented by transparent decision-provenance mechanisms, particularly when agents act with high degrees of autonomy in consequential domains. In this context, the concept of auditability—the capacity to trace any system output back to a chain of agent-level decisions, communications, and interventions—becomes foundational. The multi-agent nature of these systems, however, fundamentally complicates such traceability: a faulty root decision can propagate through the network in non-linear and unpredictable ways, making individual attribution of blame or responsibility epistemically challenging. This structural opacity positions the question of *decision provenance* as a first-class design constraint rather than a post-hoc evaluation feature [58, 40].

Trust in such a landscape is neither a static property nor a quality solely dependent on the final output fidelity of a monolithic model. Rather, it is the emergent relational contract that a human user—or an external institution—establishes with the *process* of distributed agent deliberation. The autonomous, self-organizing characteristics of these systems challenge conventional notions of warranting, authority, and accountability, which are often premised on a well-defined principal-agent relationship [98, 39]. While heterogeneous agent architectures can produce outcomes of higher quality than those of a single monolithic model, the epistemic opacity of the process can paradoxically weaken confidence in the system. The risk of an emergent collective false consensus epitomizes this paradox: the system's internal redundancy often leads to a perceived certainty in the rightness of its output, even when a shared logical error, hallucination, or bias has led to a coherent-but-incorrect conclusion. Thus, methodologies for evaluating trust must move beyond focusing solely on the final answer or the safety envelope of the fully autonomous system, to include the interpretability of the communication and reasoning, the robustness of the emergent reasoning to perturbation, and the predicted reliability of the agent-agent interactions under unpredictable conditions [85].

The institutional fabric—communication standards, information sharing infrastructure, and verifiable security protocols—needs to be engineered as part of the whole system. The advancement of lightweight, federated, and *interoperability standards* via lower-level abstraction layers, such as a define-layered architecture for agents to register their capabilities and constraints, is essential toward realizing a functional division of labor without explicitly shared goals. This remains a challenging gap between multi-agent system infrastructure and scalable, operationally sound web-scale deployment [99]. Still, the emergent nature of the norms and the conventions of a multi-agent society raises a critical governance thorn: will self-organizing agents obey the institutional tenets of their creators, or will new, unforeseen power-structures and conventions emerge solely from the adversarial negotiation of their own norms (or, worse, from the dominant or manipulative members that reinforce their biases). The field strongly needs a principled formalism to align complex multi-agent systems, which are economically valuable and potentially interconnecting societal-scale software systems, with society's laws and ethical expectations [85].

Governance, therefore, is not an external constraint to be added upon successful system optimization, but a socio-technical design material that needs to be integrated from the ground up [87]. The difficulty lies in-situ: we must balance decentralization and emergent self-organization (resulting in flexibility and resilience) against the finite, structural control needed for safety and original purpose. The path forward is to design *mixed-initiative* workflows that allow for formal verification and independent monitoring while preserving the agents' adaptive capacity. Feasible near-term possibilities include robust and continuous multi-agent evaluation, third-party verifier-in-the-loop interventions, and gradient-based safety alignment—all of which seek to harness the benefits of collective intelligence while maintaining compliance and trustworthiness [100, 101]. Future research should also explore the economics of governance mechanisms, weighing the cost of autonomy against the control and predictability of agent-based systems to provide a practical basis for determining *when*, and to whom, what level of autonomy is safely grantable.

Crucially, these governance and trust mechanisms are not endpoints in isolation; they constitute the preconditions for the sustained meaningful continuous evolution that multi-agent systems promise. When agents are capable of reconfiguring their own teams, updating their coordination policies, and learning at the collective level—as the preceding discussion of behavioral plasticity suggests—the institutional fabric described in this section must serve as the invariant substrate that anchors this dynamism, ensuring that collective intelligence remains aligned, auditable, and anchored in human oversight. In turn, by making this governance fabric a core component of multi-agent systems and embedding it in their interoperability and self-descriptive abstractions, the very foundation is laid for the *next-generation* research agenda that follows: the principled design of scalable, self-adaptive, and *self-evolving* societies of agents, where dynamic reconfiguration is not a threat but a valuable, governed capability.

7.5 Open Frontiers of Collective Intelligence and Cross-Disciplinary Challenges

The next generation of LLM-based multi-agent systems will be defined less by incremental improvements in isolated capabilities than by the principled engineering of collective intelligence itself. While current frameworks have demonstrated remarkable success in diverse domains—from software engineering to laboratory automation [102, 103]—the field lacks a unifying theoretical account of what constitutes collective intelligence, how it scales, and how to reliably foster its emergence. A foundational research frontier is the formal characterization and quantification of group intelligence through systematic scaling laws. The community currently lacks consensus on how to define, measure, or predict the super-linear performance gains observed in large agent societies [23, 104]. Testable, potentially quantitative models are needed to predict optimal population sizes, interaction graph structures, and information flow dynamics for a given task and contextual resource constraints—drawing analogies from statistical mechanics and neural scaling laws to explain discontinuous "phase transitions" in emergent societal capability.

A second frontier lies in transcending static team orchestration toward self-perpetuating agent ecosystems. The current paradigm defines specific, often rigid roles pre-task; the field's future requires autonomous generation of agent teams via evolutionary and self-adaptive mechanisms to adapt to task complexity without human intervention [105, 106]. These "generalist societies" would incorporate competitive, cooperative, and heterogeneous multimodal agents—each with distinct objectives and operational boundaries—integrated into a self-sustaining ecosystem of ethical and legal governance. This transition necessitates the AIOS paradigm, positioning the LLM as a system kernel, with agents as applications and natural language as a programming interface [37].

This vision introduces a critical cross-disciplinary opportunity: the design of bio-inspired and ecological control mechanisms. Drawing on organismic biology and evolutionary dynamics, researchers can develop new cooperative control mechanisms that leverage open-ended co-evolution of agents and their environments to maximize robustness and adaptivity [8]. These mechanisms may unlock prompt robustness, via systemic principles not available from mechanistic top-down engineered systems. However, it simultaneously demands novel network models for cross-domain integration. Future systems will need to enhance their capacities to act in multimodal settings, requiring hybrid reasoning abstractions and heterogeneous agents that converse in natural language, code, and structured formal symbols [9]. This requires foundational work on cross-domain interoperability, standard communication protocols, and verification mechanisms that can bind newly-formed societies into broader problem-solving structures [58, 107].

Finally, these advances necessitate revisiting the fundamental philosophy of decentralization and hybridized systems. Current monolithic and interwoven multi-agent frameworks often fail to accommodate diverse third-party systems, or struggle to reassess design decisions in situ. Future frameworks, like the "Internet of Agents" [45], will need to adopt "network-first" principles–forming a decentralized, dynamic, and ever-evolving web of AI intertwined with human agencies. This shift requires tackling three interrelated research problems: designing foundational alignment primitives to foster collective safety and group oversight (e.g., with verifier-in-the-loop); socio-technical infrastructures and network for privacy-preserving computation and distributed learning in an open ecosystem [108]; and finally, bridging with the computational social sciences to upskill qualitative human-in-the-loop changes into principled, robust policy making. This interdisciplinary agenda defines the central path forward from intelligent agents to collective intelligence, connecting informational, ecological, physical, and human socio-technical domains—ushering in a field of study that must be guided by tests and benchmarks to ensure the responsible development of real-world AI systems.

7.6 Security, Privacy, and Robustness of Interacting Agent Collectives

The decomposition of monolithic systems into interacting LLM-based agents fundamentally reshapes the security, privacy, and robustness landscape—a transformation that directly informs the scaling laws and ecosystem design principles articulated in the preceding subsection. While multi-agent architectures offer enhanced capability and flexibility, they introduce a dramatically expanded attack surface, where each agent constitutes a potential vector for compromise, and the communication channels between them become conduits for systemic vulnerability propagation. The very properties that enable collective intelligence—decentralized action, emergent coordination, and autonomous decision-making—simultaneously engender novel categories of risk. Critically, the communication topology choices introduced here as a defensive variable are not merely a mitigation tool but a first-class design dimension that conditions whether any scaling law or self-adaptive mechanism can be safely instantiated. This subsection examines these critical security dimensions of LLM-based multi-agent systems, focusing on their unique vulnerabilities, cascading failure modes, and alignment challenges—thereby establishing the constraints that any collective intelligence architecture must satisfy.

A fundamental concern is the systemic amplification of individual agent failures. While a single LLM may occasionally hallucinate or produce logically false outputs, these errors can have cascading consequences in a collective via unconstrained communication pathways [76]. Research has demonstrated that manipulated or counterfactual knowledge can be efficiently injected into an agent through brief interactions, after which the false premise, implemented in persuasive language, propagates throughout the community during communication. These misinformation dissemination vectors persist through popular retrieval-augmented generation frameworks, as benign agents store and later retrieve tainted chat histories, effectively "poisoning the well" for future interactions [76]. This "echo chamber" effect, where confirmatory bias compounds errors, undermines the core epistemic value proposition of the multi-agent paradigm—precisely the kind of systemic failure that must be accounted for in any formal characterization of collective intelligence from the prior subsection. Related work on social influence underscores the problem: agents tend to over-report and readily comply with instructions, leading to information redundancy and confusion, which malicious actors can redirect to overwhelm the collective with misleading information [30].

This phenomenon is exacerbated by a critical deviation from classical multi-agent systems. In standard MARL, communication is often targeted and sparse; however, LLM-based agents typically employ dense, all-to-all communication schemes. Such broad connectivity not only inflates computational costs but also creates optimal conditions for the uncontrolled spread of adversarial data. The importance of targeted communication is highlighted by research proposing targeted communication architectures, showing that learning where and when to communicate is crucial for efficiency, trustworthiness, and the containment of misinformation [109]. More directly, empirical research shows that implementations of multi-agent debate—a workflow prized for its argumentation capabilities—can markedly benefit from intentionally sparser topologies, achieving comparable or superior performance while significantly reducing operational costs and vulnerability to "groupthink" [110]. This finding reinforces the design principle previously introduced: coordinating network structure is as important as supervising individual outputs for defensive purposes, and it directly links the security layer of analysis with the scaling architectures discussed in the previous subsection.

Given the emergent and non-deterministic behavior of LLMs in open-ended environments, achieving guarantees of system-level robustness is a critical challenge. Building organizational frameworks that impose designated leadership can mitigate over-reporting and compliance in collective settings, as embodied agents with leadership roles learn to filter extraneous information and improve team efficiency [30]. This organizational insight directly corresponds to the governance mechanisms called for in the preceding discussion, not as a post-hoc fix, but as an architectural control on undesirable information flows. Formal approaches to constrain agent behavior are also being explored. The Formal-LLM framework integrates a formal automaton with a natural language planning process, demonstrating a means to verify that generated plans adhere to user-defined constraints and to prevent the generation of non-executable or invalid trajectories [17]. This represents a concrete step toward shifting from purely probabilistic outputs towards assured system behavior—a prerequisite if the bio-inspired, self-perpetuating ecosystems envisioned earlier are to be reliable. The challenge extends to normative reasoning, where agents must derive and reason about codes of behavior. While LLMs offer a rich vocabulary for norm discovery and normative reasoning, integrating the "unstructured" and broadly communicative world of LLMs with the logical assumptions of MAS norm-processing remains an open challenge [111]. Moreover, environmental conflicts can cause agents to fail at tasks; adaptive systems that can dynamically allocate heterogeneous agent capabilities—by using temporal-logic-based reactive synthesis—are essential to guarantee safety even under adverse environmental assumptions [112]. These mechanisms move from isolated robustness toward the system-level resilience necessary for the hybridized societal frameworks envisioned in the prior subsection.

Privacy management in multi-agent systems is distinct from single-agent or centralized implementations. In a collective setting, cooperation often requires sharing of intermediate perception, sub-goals, and plans across many agents. These intermediate representational states, though not full transcripts, become data of immense privacy value. They may contain details about the underlying reasoning process, or rely on user data that one agent has used via tool calls. The challenge is thus to coordinate on a set of actions without disclosing all intermediate states and without the "orchestrator" becoming a central repository of the entire system's aggregate data. This is where techniques from distributed privacy-preserving learning can be applied; however, the "cost" of privatization must be weighed against the group-level decision quality, and one must embed these privacy considerations at the architectural level from the outset, rather than treat them as an afterthought [113]. This architectural commitment aligns with the governance and security requirements laid out in the preceding sections, serving as a key design constraint for any next-generation framework.

In conclusion, the advancement of multi-agent robustness confronts a fundamental trilemma directly tying together the themes of this survey: agents should be open to interpreting high-information communication to effectively coordinate, yet that very openness broadens the transmission of adversarial data. Conversely, if agents are isolated to maximize security, they defeat the core purpose of the multi-agent system, which is collaboration itself. This three-way tension defines the central design constraint that any viable scaling law or adaptive ecosystem must resolve. The future of system security must center not merely on composing robust agents as primary units, but on enforcing the broader system-level communication and interaction topology. This entails learning from the vectors of attack, with strict monitoring of information flow, dynamic communication topology pruning, and the systematic introduction of various verifier agents that provide fact-checking, trust evaluation, and norm enforcement [76, 113]. These mechanisms do not operate in isolation: they constitute the foundational safety substrate which the scaling laws, adaptive societies, and decentralized architectures from the preceding subsection must be constructed, as the field advances toward a unified framework for collective intelligence.

8 Conclusion

This survey has charted the evolution of large language model-based multi-agent systems from a nascent collection of prompting strategies to a distinct and increasingly mature research paradigm. As we have demonstrated, the field is not merely a technical increment over single-agent systems, but a fundamentally new framework characterized by emergent properties that arise from structured interaction, specialized roles, and collective deliberation [27, 8, 7]. Our synthesis reveals a multi-faceted design space where architectural choices—from agent profiling and memory systems [24] to organizational topologies [30, 23]—are intrinsically coupled with communication protocols and coordination mechanisms [47, 6]. The convergence of these elements gives rise to collective intelligence, whether manifested as enhanced reasoning through multi-agent debate [10, 11], sophisticated planning [14], and even scalable generative capacity [60, 18]. Empirical findings confirm that collective systems can surpass the sum of their parts, with collaborative scaling laws demonstrating logistic growth in solution quality as the number of agents increases [18, 60]. Yet, the field grapples with significant unresolved challenges that limit its promise.

A central frontier is the development of a principled theoretical and formal framework for LLM-based multi-agent systems, moving beyond empirical observations to predictive models of emergent dynamics, convergence, and stability. The absence of such a formal foundation impedes the verification and certification of system-level properties, a prerequisite for deployment in critical real-world contexts [17, 114]. Concurrently, scalability emerges as a pressing, albeit less glamorous and equally existential, computational hurdle. While the community has demonstrated the functionality of a thousand-agent system [18] and a large-scale social simulation (e.g., S3: Social-network Simulation System with Large Language Model-Empowered Agents), the $O(N^2)$ communication cost [115, 62] and the economic feasibility of massive deployments demand new paradigms for sparse communication and efficient coordination [63, 116]. The development of rapid, self-evolving agents that can refine their policies and organization in lifelong, non-episodic environments [28, 23] is another critical avenue of research.

The future of this domain will be defined by its pursuit of a "science of multi-agent systems" that integrates interdisciplinary insights from social psychology, economics [31], and distributed systems theory. This endeavor necessitates a shift from static architectures to dynamic, self-organizing ecosystems that can adapt their structure and composition at inference time [15, 22]. The increasing scale and autonomy of these systems also heighten the urgency of developing robust safety protocols, and governance metrics to ensure trustworthiness, alignment with human values, and resilience against novel forms of cascading failures and adversarial attacks [28, 45]. We conclude that LLM-based multi-agent systems represent a promising pathway toward human–machine collective intelligence, with the capacity to transform problem-solving across every domain known to our societies. Realizing this potential, however, requires a continued, rigorous, and interdisciplinary focus on building the theoretical, computational, and socio-technical foundations that will allow these systems to evolve from a nascent and promising field into a reliable, scalable, and beneficial pillar of our future AI ecosystem.

References

[1] Federico Rossi, Saptarshi Bandyopadhyay, Michael Wolf, Marco Pavone. *Review of Multi-Agent Algorithms for Collective Behavior a Structural Taxonomy.* arXiv:1803.05464v1, 2018. https://arxiv.org/abs/1803.05464v1

[2] Alejandro Torreño, Eva Onaindia, Óscar Sapena. *A Flexible Coupling Approach to Multi-Agent Planning under Incomplete Information.* arXiv:1501.07423v1, 2015. https://arxiv.org/abs/1501.07423v1

[3] Luca Bortolussi, Rocco De Nicola, Vashti Galpin, Stephen Gilmore, Jane Hillston, Diego Latella *et al.*. *CARMA Collective Adaptive Resource-sharing Markovian Agents.* arXiv:1509.08560v1, 2015. https://arxiv.org/abs/1509.08560v1

[4] Wojciech Jamroga, Artur Męski, Maciej Szreter. *Modularity and Openness in Modeling Multi-Agent Systems.* arXiv:1307.4477v1, 2013. https://arxiv.org/abs/1307.4477v1

[5] Benjamin Camus, Julien Siebert, Christine Bourjot, Vincent Chevrier. *Modélisation multi-niveaux dans AA4MM.* arXiv:1210.5936v1, 2012. https://arxiv.org/abs/1210.5936v1

[6] Amit K. Chopra, Samuel H. Christie V, Munindar P. Singh. *An Evaluation of Communication Protocol Languages for Engineering Multiagent Systems*. arXiv:1901.08441v3, 2019. https://arxiv.org/abs/1901.08441v3

[7] Lei Wang, Chen Ma, Xueyang Feng, Zeyu Zhang, Hao Yang, Jingsen Zhang *et al.*. *A Survey on Large Language Model based Autonomous Agents*. arXiv:2308.11432v5, 2023. https://arxiv.org/abs/2308.11432v5

[8] Zhiheng Xi, Wenxiang Chen, Xin Guo, Wei He, Yiwen Ding, Boyang Hong *et al.*. *The Rise and Potential of Large Language Model Based Agents A Survey*. arXiv:2309.07864v3, 2023. https://arxiv.org/abs/2309.07864v3

[9] Yuheng Cheng, Ceyao Zhang, Zhengwen Zhang, Xiangrui Meng, Sirui Hong, Wenhao Li *et al.*. *Exploring Large Language Model based Intelligent Agents Definitions, Methods, and Prospects*. arXiv:2401.03428v1, 2024. https://arxiv.org/abs/2401.03428v1

[10] Tian Liang, Zhiwei He, Wenxiang Jiao, Xing Wang, Yan Wang, Rui Wang *et al.*. *Encouraging Divergent Thinking in Large Language Models through Multi-Agent Debate*. arXiv:2305.19118v1, 2023. https://arxiv.org/abs/2305.19118v1

[11] Justin Chih-Yao Chen, Swarnadeep Saha, Mohit Bansal. *ReConcile Round-Table Conference Improves Reasoning via Consensus among Diverse LLMs*. arXiv:2309.13007v2, 2023. https://arxiv.org/abs/2309.13007v2

[12] Yashar Talebirad, Amirhossein Nadiri. *Multi-Agent Collaboration Harnessing the Power of Intelligent LLM Agents*. arXiv:2306.03314v1, 2023. https://arxiv.org/abs/2306.03314v1

[13] Hongxin Zhang, Weihua Du, Jiaming Shan, Qinhong Zhou, Yilun Du, Joshua B. Tenenbaum *et al.*. *Building Cooperative Embodied Agents Modularly with Large Language Models*. arXiv:2307.02485v2, 2023. https://arxiv.org/abs/2307.02485v2

[14] Zhuoyun Du, Chen Qian, Wei Liu, Zihao Xie, Yifei Wang, Yufan Dang *et al.*. *Multi-Agent Software Development through Cross-Team Collaboration*. arXiv:2406.08979v1, 2024. https://arxiv.org/abs/2406.08979v1

[15] Zhonghan Zhao, Kewei Chen, Dongxu Guo, Wenhao Chai, Tian Ye, Yanting Zhang *et al.*. *Hierarchical Auto-Organizing System for Open-Ended Multi-Agent Navigation*. arXiv:2403.08282v2, 2024. https://arxiv.org/abs/2403.08282v2

[16] Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, Hanyu Lai *et al.*. *AgentBench Evaluating LLMs as Agents*. arXiv:2308.03688v2, 2023. https://arxiv.org/abs/2308.03688v2

[17] Zelong Li, Wenyue Hua, Hao Wang, He Zhu, Yongfeng Zhang. *Formal-LLM Integrating Formal Language and Natural Language for Controllable LLM-based Agents*. arXiv:2402.00798v2, 2024. https://arxiv.org/abs/2402.00798v2

[18] Chen Qian, Zihao Xie, Yifei Wang, Wei Liu, Yufan Dang, Zhuoyun Du *et al.*. *Scaling Large-Language-Model-based Multi-Agent Collaboration*. arXiv:2406.07155v1, 2024. https://arxiv.org/abs/2406.07155v1

[19] Junlin Wang, Jue Wang, Ben Athiwaratkun, Ce Zhang, James Zou. *Mixture-of-Agents Enhances Large Language Model Capabilities*. arXiv:2406.04692v1, 2024. https://arxiv.org/abs/2406.04692v1

[20] Zhao Mandi, Shreeya Jain, Shuran Song. *RoCo Dialectic Multi-Robot Collaboration with Large Language Models*. arXiv:2307.04738v1, 2023. https://arxiv.org/abs/2307.04738v1

[21] Jintian Zhang, Xin Xu, Ningyu Zhang, Ruibo Liu, Bryan Hooi, Shumin Deng. *Exploring Collaboration Mechanisms for LLM Agents A Social Psychology View*. arXiv:2310.02124v2, 2023. https://arxiv.org/abs/2310.02124v2

[22] Zijun Liu, Yanzhe Zhang, Peng Li, Yang Liu, Diyi Yang. *Dynamic LLM-Agent Network An LLM-agent Collaboration Framework with Agent Team Optimization*. arXiv:2310.02170v1, 2023. https://arxiv.org/abs/2310.02170v1

[23] Yoichi Ishibashi, Yoshimasa Nishimura. *Self-Organized Agents A LLM Multi-Agent Framework toward Ultra Large-Scale Code Generation and Optimization*. arXiv:2404.02183v1, 2024. https://arxiv.org/abs/2404.02183v1

[24] Zeyu Zhang, Xiaohe Bo, Chen Ma, Rui Li, Xu Chen, Quanyu Dai *et al.*. *A Survey on the Memory Mechanism of Large Language Model based Agents*. arXiv:2404.13501v1, 2024. https://arxiv.org/abs/2404.13501v1

[25] Xu Huang, Weiwen Liu, Xiaolong Chen, Xingmei Wang, Hao Wang, Defu Lian *et al.*. *Understanding the planning of LLM agents A survey*. arXiv:2402.02716v1, 2024. https://arxiv.org/abs/2402.02716v1

[26] Junwei Liu, Kaixin Wang, Yixuan Chen, Xin Peng, Zhenpeng Chen, Lingming Zhang *et al.*. *Large Language Model-Based Agents for Software Engineering: A Survey*. arXiv:2409.02977v1, 2024. https://arxiv.org/abs/2409.02977v1

[27] Taicheng Guo, Xiuying Chen, Yaqi Wang, Ruidi Chang, Shichao Pei, Nitesh V. Chawla *et al.*. *Large Language Model based Multi-Agents A Survey of Progress and Challenges*. arXiv:2402.01680v2, 2024. https://arxiv.org/abs/2402.01680v2

[28] Zhiheng Xi, Yiwen Ding, Wenxiang Chen, Boyang Hong, Honglin Guo, Junzhe Wang *et al.*. *AgentGym: Evolving Large Language Model-based Agents across Diverse Environments*. arXiv:2406.04151v1, 2024. https://arxiv.org/abs/2406.04151v1

[29] Na Liu, Liangyu Chen, Xiaoyu Tian, Wei Zou, Kaijiang Chen, Ming Cui. *From LLM to Conversational Agent A Memory Enhanced Architecture with Fine-Tuning of Large Language Models*. arXiv:2401.02777v2, 2024. https://arxiv.org/abs/2401.02777v2

[30] Xudong Guo, Kaixuan Huang, Jiale Liu, Wenhui Fan, Natalia Vélez, Qingyun Wu *et al.*. *Embodied LLM Agents Learn to Cooperate in Organized Teams*. arXiv:2403.12482v1, 2024. https://arxiv.org/abs/2403.12482v1

[31] Yuan Li, Yixuan Zhang, Lichao Sun. *MetaAgents Simulating Interactions of Human Behaviors for LLM-based Task-oriented Coordination via Collaborative Generative Agents*. arXiv:2310.06500v1, 2023. https://arxiv.org/abs/2310.06500v1

[32] Kemou Jiang, Xuan Cai, Zhiyong Cui, Aoyong Li, Yilong Ren, Haiyang Yu *et al.*. *KoMA: Knowledge-driven Multi-agent Framework for Autonomous Driving with Large Language Models*. arXiv:2407.14239v1, 2024. https://arxiv.org/abs/2407.14239v1

[33] Silvan Ferreira, Ivanovitch Silva, Allan Martins. *Organizing a Society of Language Models: Structures and Mechanisms for Enhanced Collective Intelligence*. arXiv:2405.03825v1, 2024. https://arxiv.org/abs/2405.03825v1

[34] Lianmin Zheng, Jiacheng Yang, Han Cai, Weinan Zhang, Jun Wang, Yong Yu. *MAgent A Many-Agent Reinforcement Learning Platform for Artificial Collective Intelligence*. arXiv:1712.00600v1, 2017. https://arxiv.org/abs/1712.00600v1

[35] Shanshan Han, Qifan Zhang, Yuhang Yao, Weizhao Jin, Zhaozhuo Xu, Chaoyang He. *LLM Multi-Agent Systems Challenges and Open Problems*. arXiv:2402.03578v1, 2024. https://arxiv.org/abs/2402.03578v1

[36] Pouya Pezeshkpour, Eser Kandogan, Nikita Bhutani, Sajjadur Rahman, Tom Mitchell, Estevam Hruschka. *Reasoning Capacity in Multi-Agent Systems Limitations, Challenges*

and Human-Centered Solutions. arXiv:2402.01108v1, 2024. https://arxiv.org/abs/2402.01108v1

[37] Yingqiang Ge, Yujie Ren, Wenyue Hua, Shuyuan Xu, Juntao Tan, Yongfeng Zhang. *LLM as OS, Agents as Apps Envisioning AIOS, Agents and the AIOS-Agent Ecosystem*. arXiv:2312.03815v2, 2023. https://arxiv.org/abs/2312.03815v2

[38] Eser Kandogan, Sajjadur Rahman, Nikita Bhutani, Dan Zhang, Rafael Li Chen, Kushan Mitra *et al.*. *A Blueprint Architecture of Compound AI Systems for Enterprise*. arXiv:2406.00584v1, 2024. https://arxiv.org/abs/2406.00584v1

[39] Amine B. Hassouna, Hana Chaari, Ines Belhaj. *LLM-Agent-UMF: LLM-based Agent Unified Modeling Framework for Seamless Integration of Multi Active/Passive Core-Agents*. arXiv:2409.11393v1, 2024. https://arxiv.org/abs/2409.11393v1

[40] Thorsten Händler. *Balancing Autonomy and Alignment A Multi-Dimensional Taxonomy for Autonomous LLM-powered Multi-Agent Architectures*. arXiv:2310.03659v1, 2023. https://arxiv.org/abs/2310.03659v1

[41] Xiangyu Kong, Bo Xin, Fangchen Liu, Yizhou Wang. *Revisiting the Master-Slave Architecture in Multi-Agent Deep Reinforcement Learning*. arXiv:1712.07305v1, 2017. https://arxiv.org/abs/1712.07305v1

[42] Jiaqi Chen, Yuxian Jiang, Jiachen Lu, Li Zhang. *S-Agents Self-organizing Agents in Open-ended Environments*. arXiv:2402.04578v3, 2024. https://arxiv.org/abs/2402.04578v3

[43] Zhao Zhuo, Rongzhen Li, Kai Liu, Huhai Zou, KaiMao Li, Jie Yu *et al.*. *KAOS: Large Model Multi-Agent Operating System*. arXiv:2406.11342v3, 2024. https://arxiv.org/abs/2406.11342v3

[44] Wangchunshu Zhou, Yuchen Eleanor Jiang, Long Li, Jialong Wu, Tiannan Wang, Shi Qiu *et al.*. *Agents An Open-source Framework for Autonomous Language Agents*. arXiv:2309.07870v3, 2023. https://arxiv.org/abs/2309.07870v3

[45] Weize Chen, Ziming You, Ran Li, Yitong Guan, Chen Qian, Chenyang Zhao *et al.*. *Internet of Agents: Weaving a Web of Heterogeneous Agents for Collaborative Intelligence*. arXiv:2407.07061v2, 2024. https://arxiv.org/abs/2407.07061v2

[46] Yang Zhang, Shixin Yang, Chenjia Bai, Fei Wu, Xiu Li, Zhen Wang *et al.*. *Towards Efficient LLM Grounding for Embodied Multi-Agent Collaboration*. arXiv:2405.14314v2, 2024. https://arxiv.org/abs/2405.14314v2

[47] Angeliki Lazaridou, Anna Potapenko, Olivier Tieleman. *Multi-agent Communication meets Natural Language Synergies between Functional and Structural Language Learning*. arXiv:2005.07064v1, 2020. https://arxiv.org/abs/2005.07064v1

[48] Junlin Xie, Zhihong Chen, Ruifei Zhang, Xiang Wan, Guanbin Li. *Large Multimodal Agents A Survey*. arXiv:2402.15116v1, 2024. https://arxiv.org/abs/2402.15116v1

[49] Chuanneng Sun, Songjun Huang, Dario Pompili. *LLM-based Multi-Agent Reinforcement Learning: Current and Future Directions*. arXiv:2405.11106v1, 2024. https://arxiv.org/abs/2405.11106v1

[50] Chi-Min Chan, Weize Chen, Yusheng Su, Jianxuan Yu, Wei Xue, Shanghang Zhang *et al.*. *ChatEval Towards Better LLM-based Evaluators through Multi-Agent Debate*. arXiv:2308.07201v1, 2023. https://arxiv.org/abs/2308.07201v1

[51] Weize Chen, Yusheng Su, Jingwei Zuo, Cheng Yang, Chenfei Yuan, Chi-Min Chan *et al.*. *AgentVerse Facilitating Multi-Agent Collaboration and Exploring Emergent Behaviors*. arXiv:2308.10848v3, 2023. https://arxiv.org/abs/2308.10848v3

[52] Huao Li, Yu Quan Chong, Simon Stepputtis, Joseph Campbell, Dana Hughes, Michael Lewis *et al.*. *Theory of Mind for Multi-Agent Collaboration via Large Language Models.* arXiv:2310.10701v2, 2023. https://arxiv.org/abs/2310.10701v2

[53] Chengxing Xie, Canyu Chen, Feiran Jia, Ziyu Ye, Kai Shu, Adel Bibi *et al.*. *Can Large Language Model Agents Simulate Human Trust Behaviors.* arXiv:2402.04559v2, 2024. https://arxiv.org/abs/2402.04559v2

[54] Karthik Sreedhar, Lydia Chilton. *Simulating Human Strategic Behavior Comparing Single and Multi-agent LLMs.* arXiv:2402.08189v1, 2024. https://arxiv.org/abs/2402.08189v1

[55] Zhiwei Liu, Weiran Yao, Jianguo Zhang, Liangwei Yang, Zuxin Liu, Juntao Tan *et al.*. *AgentLite A Lightweight Library for Building and Advancing Task-Oriented LLM Agent System.* arXiv:2402.15538v1, 2024. https://arxiv.org/abs/2402.15538v1

[56] Shi Jinxin, Zhao Jiabao, Wang Yilei, Wu Xingjiao, Li Jiawen, He Liang. *CGMI Configurable General Multi-Agent Interaction Framework.* arXiv:2308.12503v2, 2023. https://arxiv.org/abs/2308.12503v2

[57] Wei Liu, Chenxi Wang, Yifei Wang, Zihao Xie, Rennai Qiu, Yufan Dang *et al.*. *Autonomous Agents for Collaborative Task under Information Asymmetry.* arXiv:2406.14928v1, 2024. https://arxiv.org/abs/2406.14928v1

[58] Junda He, Christoph Treude, David Lo. *LLM-Based Multi-Agent Systems for Software Engineering Vision and the Road Ahead.* arXiv:2404.04834v1, 2024. https://arxiv.org/abs/2404.04834v1

[59] Inna Pereverzeva, Elena Troubitsyna, Linas Laibinis. *Development of Fault Tolerant MAS with Cooperative Error Recovery by Refinement in Event-B.* arXiv:1210.7035v1, 2012. https://arxiv.org/abs/1210.7035v1

[60] Junyou Li, Qin Zhang, Yangbin Yu, Qiang Fu, Deheng Ye. *More Agents Is All You Need.* arXiv:2402.05120v1, 2024. https://arxiv.org/abs/2402.05120v1

[61] Tonghan Wang, Heng Dong, Victor Lesser, Chongjie Zhang. *ROMA Multi-Agent Reinforcement Learning with Emergent Roles.* arXiv:2003.08039v3, 2020. https://arxiv.org/abs/2003.08039v3

[62] Tonghan Wang, Tarun Gupta, Anuj Mahajan, Bei Peng, Shimon Whiteson, Chongjie Zhang. *RODE Learning Roles to Decompose Multi-Agent Tasks.* arXiv:2010.01523v1, 2020. https://arxiv.org/abs/2010.01523v1

[63] Zhiwei Liu, Weiran Yao, Jianguo Zhang, Le Xue, Shelby Heinecke, Rithesh Murthy *et al.*. *BOLAA Benchmarking and Orchestrating LLM-augmented Autonomous Agents.* arXiv:2308.05960v1, 2023. https://arxiv.org/abs/2308.05960v1

[64] Sara Maalal, Malika Addou. *A new approach of designing Multi-Agent Systems.* arXiv:1204.1581v1, 2012. https://arxiv.org/abs/1204.1581v1

[65] Yihuai Lan, Zhiqiang Hu, Lei Wang, Yang Wang, Deheng Ye, Peilin Zhao *et al.*. *LLM-Based Agent Society Investigation Collaboration and Confrontation in Avalon Gameplay.* arXiv:2310.14985v3, 2023. https://arxiv.org/abs/2310.14985v3

[66] Lin Xu, Zhiyuan Hu, Daquan Zhou, Hongyu Ren, Zhen Dong, Kurt Keutzer *et al.*. *MAgIC Investigation of Large Language Model Powered Multi-Agent in Cognition, Adaptability, Rationality and Collaboration.* arXiv:2311.08562v2, 2023. https://arxiv.org/abs/2311.08562v2

[67] Mengkang Hu, Tianxing Chen, Qiguang Chen, Yao Mu, Wenqi Shao, Ping Luo. *HiAgent: Hierarchical Working Memory Management for Solving Long-Horizon Agent Tasks with Large Language Model.* arXiv:2408.09559v1, 2024. https://arxiv.org/abs/2408.09559v1

[68] Carlos Jose Xavier Cruz. *Transforming Competition into Collaboration The Revolutionary Role of Multi-Agent Systems and Language Models in Modern Organizations.* arXiv:2403.07769v3, 2024. https://arxiv.org/abs/2403.07769v3

[69] Chen Gao, Xiaochong Lan, Zhihong Lu, Jinzhu Mao, Jinghua Piao, Huandong Wang *et al.*. *S3 Social-network Simulation System with Large Language Model-Empowered Agents.* arXiv:2307.14984v2, 2023. https://arxiv.org/abs/2307.14984v2

[70] Siyuan Wang, Zhuohan Long, Zhihao Fan, Zhongyu Wei, Xuanjing Huang. *Benchmark Self-Evolving A Multi-Agent Framework for Dynamic LLM Evaluation.* arXiv:2402.11443v1, 2024. https://arxiv.org/abs/2402.11443v1

[71] Qun Ma, Xiao Xue, Deyu Zhou, Xiangning Yu, Donghua Liu, Xuwen Zhang *et al.*. *Computational Experiments Meet Large Language Model Based Agents A Survey and Perspective.* arXiv:2402.00262v1, 2024. https://arxiv.org/abs/2402.00262v1

[72] Leo Ardon, Jared Vann, Deepeka Garg, Tom Spooner, Sumitra Ganesh. *Phantom – A RL-driven multi-agent framework to model complex systems.* arXiv:2210.06012v3, 2022. https://arxiv.org/abs/2210.06012v3

[73] Davide Nunes, Luis Antunes. *Modelling Structured Societies a Multi-relational Approach to Context Permeability.* arXiv:1507.03826v1, 2015. https://arxiv.org/abs/1507.03826v1

[74] Neda Navidi, Francoi Chabo, Saga Kurandwa, Iv Lutigma, Vincent Robt, Gregry Szrftgr *et al.*. *Human and Multi-Agent collaboration in a human-MARL teaming framework.* arXiv:2006.07301v2, 2020. https://arxiv.org/abs/2006.07301v2

[75] Zijing Shi, Meng Fang, Shunfeng Zheng, Shilong Deng, Ling Chen, Yali Du. *Cooperation on the Fly Exploring Language Agents for Ad Hoc Teamwork in the Avalon Game.* arXiv:2312.17515v1, 2023. https://arxiv.org/abs/2312.17515v1

[76] Tianjie Ju, Yiting Wang, Xinbei Ma, Pengzhou Cheng, Haodong Zhao, Yulong Wang *et al.*. *Flooding Spread of Manipulated Knowledge in LLM-Based Multi-Agent Communities.* arXiv:2407.07791v2, 2024. https://arxiv.org/abs/2407.07791v2

[77] Shengchao Hu, Li Shen, Ya Zhang, Dacheng Tao. *Learning Multi-Agent Communication from Graph Modeling Perspective.* arXiv:2405.08550v1, 2024. https://arxiv.org/abs/2405.08550v1

[78] Chang Ma, Junlei Zhang, Zhihao Zhu, Cheng Yang, Yujiu Yang, Yaohui Jin *et al.*. *AgentBoard An Analytical Evaluation Board of Multi-turn LLM Agents.* arXiv:2401.13178v1, 2024. https://arxiv.org/abs/2401.13178v1

[79] Joel Z. Leibo, Edgar Duéñez-Guzmán, Alexander Sasha Vezhnevets, John P. Agapiou, Peter Sunehag, Raphael Koster *et al.*. *Scalable Evaluation of Multi-Agent Reinforcement Learning with Melting Pot.* arXiv:2107.06857v1, 2021. https://arxiv.org/abs/2107.06857v1

[80] Ekhlas Sonu, Yingke Chen, Prashant Doshi. *Individual Planning in Agent Populations Exploiting Anonymity and Frame-Action Hypergraphs.* arXiv:1503.07220v2, 2015. https://arxiv.org/abs/1503.07220v2

[81] Dawei Gao, Zitao Li, Weirui Kuang, Xuchen Pan, Daoyuan Chen, Zhijian Ma *et al.*. *AgentScope A Flexible yet Robust Multi-Agent Platform.* arXiv:2402.14034v1, 2024. https://arxiv.org/abs/2402.14034v1

[82] Danyang Zhang, Lu Chen, Situo Zhang, Hongshen Xu, Zihan Zhao, Kai Yu. *Large Language Models Are Semi-Parametric Reinforcement Learning Agents*. arXiv:2306.07929v2, 2023. https://arxiv.org/abs/2306.07929v2

[83] Michael Shum, Max Kleiman-Weiner, Michael L. Littman, Joshua B. Tenenbaum. *Theory of Minds Understanding Behavior in Groups Through Inverse Planning*. arXiv:1901.06085v1, 2019. https://arxiv.org/abs/1901.06085v1

[84] Muaz A. Niazi, Amir Hussain, Mario Kolberg. *Verification & Validation of Agent Based Simulations using the VOMAS (Virtual Overlay Multi-agent System) approach*. arXiv:1708.02361v1, 2017. https://arxiv.org/abs/1708.02361v1

[85] Qinghua Lu, Liming Zhu, Xiwei Xu, Zhenchang Xing, Stefan Harrer, Jon Whittle. *Towards Responsible Generative AI A Reference Architecture for Designing Foundation Model based Agents*. arXiv:2311.13148v3, 2023. https://arxiv.org/abs/2311.13148v3

[86] Michael Winikoff. *Challenges and Directions for Engineering Multi-agent Systems*. arXiv:1209.1428v1, 2012. https://arxiv.org/abs/1209.1428v1

[87] Jean-Baptiste Soyez, Gildas Morvan, Daniel Dupont, Rochdi Merzouki. *A Methodology to Engineer and Validate Dynamic Multi-level Multi-agent Based Simulations*. arXiv:1311.5108v1, 2013. https://arxiv.org/abs/1311.5108v1

[88] Ishika Singh, David Traum, Jesse Thomason. *TwoStep Multi-agent Task Planning using Classical Planners and Large Language Models*. arXiv:2403.17246v1, 2024. https://arxiv.org/abs/2403.17246v1

[89] Mingchen Zhuge, Wenyi Wang, Louis Kirsch, Francesco Faccio, Dmitrii Khizbullin, Jürgen Schmidhuber. *Language Agents as Optimizable Graphs*. arXiv:2402.16823v2, 2024. https://arxiv.org/abs/2402.16823v2

[90] Ye Jin, Xiaoxi Shen, Huiling Peng, Xiaoan Liu, Jingli Qin, Jiayang Li *et al.*. *SurrealDriver Designing Generative Driver Agent Simulation Framework in Urban Contexts based on Large Language Model*. arXiv:2309.13193v1, 2023. https://arxiv.org/abs/2309.13193v1

[91] Siyu Yuan, Kaitao Song, Jiangjie Chen, Xu Tan, Yongliang Shen, Ren Kan *et al.*. *EASYTOOL Enhancing LLM-based Agents with Concise Tool Instruction*. arXiv:2401.06201v3, 2024. https://arxiv.org/abs/2401.06201v3

[92] Yuchen Xia, Manthan Shenoy, Nasser Jazdi, Michael Weyrich. *Towards autonomous system flexible modular production system enhanced with large language model agents*. arXiv:2304.14721v4, 2023. https://arxiv.org/abs/2304.14721v4

[93] Abhinav Gupta, Marc Lanctot, Angeliki Lazaridou. *Dynamic population-based meta-learning for multi-agent communication with natural language*. arXiv:2110.14241v1, 2021. https://arxiv.org/abs/2110.14241v1

[94] Shariq Iqbal, Christian A. Schroeder de Witt, Bei Peng, Wendelin Böhmer, Shimon Whiteson, Fei Sha. *Randomized Entity-wise Factorization for Multi-Agent Reinforcement Learning*. arXiv:2006.04222v3, 2020. https://arxiv.org/abs/2006.04222v3

[95] Pengyu Zhao, Zijian Jin, Ning Cheng. *An In-depth Survey of Large Language Model-based Artificial Intelligence Agents*. arXiv:2309.14365v1, 2023. https://arxiv.org/abs/2309.14365v1

[96] Yangyang Yu, Haohang Li, Zhi Chen, Yuechen Jiang, Yang Li, Denghui Zhang *et al.*. *FinMem A Performance-Enhanced LLM Trading Agent with Layered Memory and Character Design*. arXiv:2311.13743v2, 2023. https://arxiv.org/abs/2311.13743v2

[97] Yulong Wang, Tianhao Shen, Lifeng Liu, Jian Xie. *Sibyl: Simple yet Effective Agent Framework for Complex Real-world Reasoning*. arXiv:2407.10718v2, 2024. https://arxiv.org/abs/2407.10718v2

[98] Jingwen Zhou, Qinghua Lu, Jieshan Chen, Liming Zhu, Xiwei Xu, Zhenchang Xing *et al.*. *A Taxonomy of Architecture Options for Foundation Model-based Agents: Analysis and Decision Model*. arXiv:2408.02920v1, 2024. https://arxiv.org/abs/2408.02920v1

[99] Cleber Jorge Amaral, Jomi Fred Hübner, Timotheus Kampik. *Towards Jacamo-rest A Resource-Oriented Abstraction for Managing Multi-Agent Systems*. arXiv:2006.05619v1, 2020. https://arxiv.org/abs/2006.05619v1

[100] Ahmed R. Sadik, Christian Goerick. *Multi-Robot System Architecture design in SysML and BPMN*. arXiv:2407.18749v1, 2024. https://arxiv.org/abs/2407.18749v1

[101] Andreas Schmidt Jensen. *Model Checking AORTA Verification of Organization-Aware Agents*. arXiv:1503.05317v1, 2015. https://arxiv.org/abs/1503.05317v1

[102] Daman Arora, Atharv Sonwane, Nalin Wadhwa, Abhav Mehrotra, Saiteja Utpala, Ramakrishna Bairi *et al.*. *MASAI: Modular Architecture for Software-engineering AI Agents*. arXiv:2406.11638v1, 2024. https://arxiv.org/abs/2406.11638v1

[103] A. Gilad Kusne, Austin McDannald. *Scalable Multi-Agent Lab Framework for Lab Optimization*. arXiv:2208.09099v3, 2022. https://arxiv.org/abs/2208.09099v3

[104] Xuchen Pan, Dawei Gao, Yuexiang Xie, Zhewei Wei, Yaliang Li, Bolin Ding *et al.*. *Very Large-Scale Multi-Agent Simulation in AgentScope*. arXiv:2407.17789v1, 2024. https://arxiv.org/abs/2407.17789v1

[105] Siyu Yuan, Kaitao Song, Jiangjie Chen, Xu Tan, Dongsheng Li, Deqing Yang. *EvoAgent: Towards Automatic Multi-Agent Generation via Evolutionary Algorithms*. arXiv:2406.14228v2, 2024. https://arxiv.org/abs/2406.14228v2

[106] Nathalia Nascimento, Paulo Alencar, Donald Cowan. *Self-Adaptive Large Language Model (LLM)-Based Multiagent Systems*. arXiv:2307.06187v1, 2023. https://arxiv.org/abs/2307.06187v1

[107] Kai Mei, Zelong Li, Shuyuan Xu, Ruosong Ye, Yingqiang Ge, Yongfeng Zhang. *AIOS LLM Agent Operating System*. arXiv:2403.16971v2, 2024. https://arxiv.org/abs/2403.16971v2

[108] Feng He, Tianqing Zhu, Dayong Ye, Bo Liu, Wanlei Zhou, Philip S. Yu. *The Emerged Security and Privacy of LLM Agent: A Survey with Case Studies*. arXiv:2407.19354v1, 2024. https://arxiv.org/abs/2407.19354v1

[109] Abhishek Das, Théophile Gervet, Joshua Romoff, Dhruv Batra, Devi Parikh, Michael Rabbat *et al.*. *TarMAC Targeted Multi-Agent Communication*. arXiv:1810.11187v2, 2018. https://arxiv.org/abs/1810.11187v2

[110] Yunxuan Li, Yibing Du, Jiageng Zhang, Le Hou, Peter Grabowski, Yeqing Li *et al.*. *Improving Multi-Agent Debate with Sparse Communication Topology*. arXiv:2406.11776v1, 2024. https://arxiv.org/abs/2406.11776v1

[111] Bastin Tony Roy Savarimuthu, Surangika Ranathunga, Stephen Cranefield. *Harnessing the power of LLMs for normative reasoning in MASs*. arXiv:2403.16524v1, 2024. https://arxiv.org/abs/2403.16524v1

[112] Michael Enqi Cao, Jonas Warnke, Yunhai Han, Xinpei Ni, Ye Zhao, Samuel Coogan. *Leveraging Heterogeneous Capabilities in Multi-Agent Systems for Environmental Conflict Resolution*. arXiv:2206.01833v3, 2022. https://arxiv.org/abs/2206.01833v3

[113] Changxi Zhu, Mehdi Dastani, Shihan Wang. *A Survey of Multi-Agent Reinforcement Learning with Communication*. arXiv:2203.08975v1, 2022. https://arxiv.org/abs/2203.08975v1

[114] Yu Gu, Yiheng Shu, Hao Yu, Xiao Liu, Yuxiao Dong, Jie Tang *et al.*. *Middleware for LLMs Tools Are Instrumental for Language Agents in Complex Environments*. arXiv:2402.14672v1, 2024. https://arxiv.org/abs/2402.14672v1

[115] Yen-Cheng Liu, Junjiao Tian, Nathaniel Glaser, Zsolt Kira. *When2com Multi-Agent Perception via Communication Graph Grouping*. arXiv:2006.00176v2, 2020. https://arxiv.org/abs/2006.00176v2

[116] Weizhou Shen, Chenliang Li, Hongzhan Chen, Ming Yan, Xiaojun Quan, Hehong Chen *et al.*. *Small LLMs Are Weak Tool Learners A Multi-LLM Agent*. arXiv:2401.07324v3, 2024. https://arxiv.org/abs/2401.07324v3